Competitive Markets for Personal Data

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June 9, 2026

Abstract

We study *competitive* data markets, in which consumers can sell their data records to information intermediaries, such as online marketplaces or advertising platforms. Using the acquired database, an intermediary designs an information policy that shapes how downstream merchants interact with consumers. Our central insight is that this competitive market may fail to maximize social welfare because of an externality. By selling her record, a consumer changes the composition of an intermediary's database. This may change how the intermediary uses the database, what merchants learn, and ultimately how they treat other consumers, even though the sold record is uninformative about them. Whether this externality leads to inefficiency depends critically on how the intermediary uses the database: generically, equilibria are inefficient if (and, under regularity conditions, only if) the intermediary's information policy is nondeterministic, and therefore may induce merchants to treat consumers with the same data differently. We discuss several solutions to this market failure, including the introduction of a data union.

JEL Classification Numbers: C72, D82, D83

Keywords: Personal Data, Data Markets, Externalities, Platforms, Data Union.

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This is a substantially revised version of a manuscript we circulated in 2024. We are thankful to Dirk Bergemann, Alessandro Bonatti, Laura Doval, Nima Haghpanah, Kevin He, Navin Kartik, Alessandro Lizzeri, Alessandro Pavan, Marco Ottaviani, Paolo Siconolfi, Philipp Strack, Laura Veldkamp, Alex Wolitzky, Kai Hao Yang, Jidong Zhou, and seminar participants at numerous universities for helpful comments. This research was supported by grants from the NSF (Galperti: SES-2149289; Perego: SES-2149315). Galperti also gratefully acknowledges financial support from UPenn's CTIC and the Warren Center for Network & Data Sciences.

1 Introduction

Across the world, lawmakers and regulators are grappling with how to govern one of the defining features of the modern economy: firms' growing reliance on consumer data. The use of consumer data can create substantial economic value, from allowing firms to personalize their products and services to supporting the development of advanced AI technologies. At the same time, it may also impose privacy costs on consumers and raise questions about who should capture the surplus generated by this critical input. As this tradeoff between the benefits and the costs of using consumer data becomes increasingly salient, many countries have begun to strengthen consumers' control over their data.[1] By making firms' access to data increasingly conditional on consumers' consent, these laws create the legal framework within which *markets for personal data* can emerge. In such markets, firms must compete to acquire data from consumers and may compensate them for the value their data helps create.

How should data markets be designed, and what outcomes should we expect from them? Despite the growing importance of these questions, the economic forces that shape these markets are not yet fully understood. This paper contributes to filling this gap by studying a canonical benchmark: a *competitive* market for personal data. Competitive markets are typically expected to align private and social incentives and thereby promote efficiency. The central insight of this paper is that, when data are used by intermediaries—as is often the case in platform-mediated markets, such as online marketplaces and advertising platforms—competition in the data market does not, by itself, guarantee efficiency. This market failure is due to an externality that is tied to how intermediaries use the data: the market is inefficient precisely when data are used to induce different treatments among consumers with the same underlying data, a potentially observable signature of the data externality.

We illustrate these forces in a two-period model featuring a heterogeneous population of consumers, a representative information intermediary (such as a digital platform), and finitely many downstream merchants. Each consumer owns a data record that, if sold to the intermediary, reveals her type and allows the intermediary to connect her with the merchants. In the first

[1] Most notably, the European Union's General Data Protection Regulation (GDPR) grants consumers the right to object to certain uses of their data, to request their data to be transferred to other firms, and to request that their data be deleted. In the past few years, similar laws have been enacted in a growing number of U.S. states, as well as in countries such as Australia, Brazil, Chile, and India.

period, consumers decide whether to sell their records, and the intermediary chooses which records to acquire, thereby forming a database. This data market is competitive in the sense that data prices are taken as given by consumers and the intermediary, and in equilibrium, clear the market. In the second period, the intermediary uses the acquired database to mediate the interaction between consumers and the merchants: it designs an information policy that shapes the merchants' behavior toward consumers. Formally, in the second period, the intermediary faces a standard information-design problem; what is distinctive is that the relevant "prior" in this problem—namely, the *composition* of the intermediary's database—is itself determined by the competitive data market in the first period. Thus, the composition of the database and the way it is used are jointly determined in equilibrium.

Our leading application is an e-commerce platform that intermediates the interaction between consumers and a single merchant. Consumers differ in their willingness to pay for the merchant's product, and the merchant uses the information provided by the platform to decide how to serve them—for example, by setting the product's price, quality, or variety. Our framework, however, applies to a broader class of information-intermediation environments in which data are acquired upstream and then used to shape an arbitrary downstream interaction. This includes, for example, online retail platforms and advertising auctions.

Our main goal is to study the properties of this data economy and, in particular, whether its equilibria are efficient, in the sense that they maximize social welfare. Our first result (Proposition 1) shows that they need not be. The reason is that a consumer's decision to sell her data may impose an externality on other consumers. By selling her record, a consumer changes the composition of the intermediary's database. This, in turn, may alter the intermediary's information policy and hence how the merchant interacts with other consumers, even if the sold record is entirely uninformative about them. These effects are not fully reflected in the consumer's private incentives. The competitive data price compensates her for the marginal value her record creates for the intermediary, but not for its indirect effect on other consumers. This wedge generates a non-pecuniary externality. We show that the absence of this data externality is necessary for efficiency and, under regularity conditions, also sufficient.

We then ask when this data externality should be expected to arise, and what economic force generates it. We find that the externality is not an inherent feature of data markets themselves; rather, it arises when the intermediary's equilibrium information policy is sensitive to the composition of the database. This sensitivity depends on a simple property of the policy itself: generically, equilibria are inefficient if—and, under regularity conditions, only if—the intermediary's information policy is *nondeterministic* (Propositions 2 and 3). A nondeterministic policy splits consumers with the same data type across different segments, thereby inducing the merchant to treat otherwise identical consumers differently. By contrast, under a deterministic policy, all consumers with the same data type are assigned to the same segment, and therefore receive the same downstream treatment.[2]

The relevance of these results is tied to the fact that nondeterministic information policies arise naturally as optimal mechanisms in a wide range of information-design environments, including price discrimination, privacy and personalized pricing, consumer profiling, and consumer-merchant matching (see, e.g., Bergemann et al., 2015; Roesler and Szentes, 2017; Strack and Yang, 2024; Elliott et al., 2025; Fainmesser et al., 2025; Condorelli and Szentes, 2026). Our results therefore raise concerns about the efficiency of competitive data markets in a broad class of applications.

To illustrate these implications, we specialize our model to one such application: the canonical third-degree price-discrimination environment of Bergemann et al. (2015). In this setting, the merchant sets the price of a product after receiving information from the intermediary about consumers' willingness to pay. We assume that the intermediary values both the revenues it obtains from selling information to the merchant and the surplus it helps create for consumers. When the intermediary puts relatively more weight on extracting the merchant's payoff, full disclosure is optimal and, thus, every equilibrium is efficient. When instead the intermediary puts relatively more weight on consumer surplus, two cases can arise: if no disclosure is the optimal information policy, the equilibrium remains efficient, since the policy is deterministic; otherwise, a nondeterministic policy is optimal, and the equilibrium is inefficient. Thus, in this canonical environment, the data market is inefficient whenever the intermediary uses consumer data in a nontrivial way, that is, when it neither fully reveals nor fully conceals consumers' types (Proposition 4).

It is useful at this point to pause and return to the central assumption of the paper—that the

[2] An important example of a deterministic policy is *full disclosure*: when in equilibrium the intermediary reveals each consumer's type to the merchant, data use is independent of the composition of the database, the data externality vanishes, and thus the equilibrium is efficient.

data market is *competitive*. Our focus on this benchmark is motivated by three considerations. First, as a descriptive matter, although many important digital intermediaries have substantial market power, others are subject to meaningful competitive pressures.[3] Second, as a policy matter, recent antitrust activity in digital markets is aimed at increasing competition, making it important to understand what competition would imply if such interventions succeed. Third, as a theoretical matter, the competitive benchmark is valuable because it helps us isolate a novel distortion, distinct from the familiar distortions associated with market power. To formalize this last point, we consider a variant of our model that is identical to the one described above except that the intermediary is a *price maker* in the data market. In that variant, we show that every equilibrium is inefficient—not because of the data externality, which is internalized, but because of a standard monopsony distortion: the intermediary marks down data prices and underacquires records (Remark 1).

The results above raise one final question: which alternative market arrangements could restore efficiency? We first consider allocating market power not to the intermediary, but to consumers themselves. In particular, we study a *data union*, an institution that manages consumers' records on their behalf, sells the resulting database to the intermediary, and redistributes the proceeds to its members. Versions of this idea have been proposed by Posner and Weyl (2018) and Bergemann et al. (2023). We show that a suitably designed data union restores efficiency, regardless of how the intermediary uses the data (Remark 2). We conclude the paper by discussing several other remedies, including data taxes, vertical integration between the merchant and the intermediary, and data prices that also depend on how the intermediary intends to use the data.

Related Literature. This paper contributes to a growing literature on the economics of digitization, data, and privacy: Acquisti et al. (2016), Bergemann and Bonatti (2019), Bergemann and Ottaviani (2021), Goldfarb and Tucker (2024), and Baley and Veldkamp (2025) provide overviews of this literature.

The closest paper to ours is Galperti et al. (2024), which characterizes an information intermediary's *demand for data*. They find that the marginal value of a data record for the intermediary need not coincide—except under full disclosure—with the average payoff generated by

[3] For instance, consider food-delivery platforms (such as DoorDash, Grubhub/Seamless, Uber Eats, and Weee!) and secondhand-apparel platforms (such as Vinted, Poshmark, and Depop).

records of that type, so that the intermediary's demand may exhibit complementarities across different types of data. Our paper embeds their demand model in a competitive data market, introducing an endogenous *supply of data* that is determined by consumers' participation decisions. This allows us to study the welfare properties of competitive data markets, the conditions under which they are inefficient, and the market designs that can restore efficiency. Importantly, in a competitive equilibrium, the price of a data record equals its marginal value for the intermediary. The complementarities identified by Galperti et al. (2024) are therefore priced in, and hence are not the source of the inefficiency we highlight.[4]

Our paper is also related to Choi et al. (2019), Ichihashi (2021), Acemoglu et al. (2022), and Bergemann et al. (2022), which study inefficiencies in markets for consumer data. At a high level, these papers differ from ours both in the mechanism that generates inefficiency and in the environments they consider. In terms of mechanism, these papers show that a consumer's decision to sell data can affect other consumers because data are correlated: one consumer's data reveals information about other consumers. We emphasize a different, and likely complementary, inefficiency, which arises instead from how intermediaries endogenously use data. To make this distinction sharp, we assume throughout that consumers' data are uncorrelated. In terms of the environment, these papers primarily focus on settings with a monopsonist data buyer rather than competitive data buyers (with the exception of Acemoglu et al. (2022)); on settings in which the data buyer uses the data itself, rather than selling information to downstream firms (with the exception of Bergemann et al. (2022)); and on settings with particular payoffs or type distributions (with the exception of Ichihashi (2021)).

Less directly, our environment is related to Kolotilin and Wolitzky (2026), in that consumers experience a form of peer effect, mediated by the intermediary's information policy. We differ in two respects. First, our supply of inputs is endogenous: consumers choose whether to participate, so the database—the prior in the downstream information-design problem—is determined in equilibrium. Second, consumers care about how their records are used, but data prices can only depend on their type, not on downstream use. Absent these two features, equilibria would be efficient, as in Kolotilin and Wolitzky (2026).[5]

[4] Indeed, our paper originated in the conjecture that, because these complementarities must be correctly priced in a competitive equilibrium, the data market would be efficient. Our results show that this conjecture is false.

[5] In an extension, Kolotilin and Wolitzky (2026) study a setting with uniform prices across consumer types, which also generates inefficiencies, though through a different mechanism.

Finally, our paper also relates to Kamenica and Lin (2025). Although the settings and the questions differ substantially, both our paper and theirs highlight the distinction between deterministic and nondeterministic mechanisms. Deterministic mechanisms are typically associated with slack obedience constraints. In their setting, this slack allows perturbing the mechanism while preserving obedience; in ours, it is the prior that is perturbed.

2 The Model

For ease of exposition, we present the model and analysis in a stylized setting with one representative platform (*it*), one merchant (*he*), and a continuum of consumers (*she*). Our results apply more broadly, as discussed at the end of this section.

Each consumer privately observes her preference type $\omega \in \Omega$ and her outside option $r \in \mathbb{R}$. We assume that Ω is finite and let $\bar{q}(\omega) > 0$ denote the mass of consumers whose preference type is ω. The outside option r is distributed according to a cumulative distribution F_ω, which has a finite mean and is differentiable with a bounded and strictly positive density f_ω. Each consumer is endowed with verifiable evidence that, if disclosed to the platform, reveals her type ω and enables the platform to mediate the interaction between this consumer and the merchant. We refer to this evidence as the consumer's *data record*.[6] By construction, a consumer's data record reveals only her own type and carries no information about the types of other consumers.

There are two periods in the model.

First Period. In the first period, consumers can sell their data records to the platform at prices $p \in \mathbb{R}^\Omega$. Both consumers and the platform take these prices as given, capturing the idea that the data market is competitive: individual consumers are atomistic, and the platform is a representative intermediary subject to competitive pressures from rival intermediaries.

More specifically, on the demand side of the data market, the platform chooses the quantity $q(\omega)$ of records of each type ω to acquire. Let $q \in \mathbb{R}_+^\Omega$ denote the composition of the resulting *database* demanded by the platform. Given a database q, the platform pays a total cost of $\sum_{\omega \in \Omega} p(\omega)q(\omega)$. On the supply side, each consumer chooses whether to sell her data record. If a type-ω consumer sells her record to the platform, she is paid the price $p(\omega)$ and moves to

[6]For example, a data record may correspond to a "cookie" containing some consumer's attributes—such as her age, gender, and location—as well as identifiers, such as her IP address or telephone number.

the second period. If she does not sell her record, she receives her outside option r and does not participate in the second period.

Second Period. In the second period, the consumers who sold their records interact with the merchant in an (arbitrary) finite game that is mediated by the platform.

More specifically, at the beginning of the second period, the composition of the database acquired by the platform becomes common knowledge, determining the merchant's belief about the consumers' types. The platform then mediates the interaction between the merchant and these consumers by acting as an *information intermediary*: it designs an information structure about consumers' types and charges the merchant a fee for access to it. After observing the signal realization, the merchant chooses an action a from a finite set A, for instance, the price, quality, or variety of the product offered to the consumer. The merchant's outside option is normalized to zero.

Together, the merchant's action a and the consumer's type ω determine the second-period payoffs: we denote the consumer's trading surplus by $u(a,\omega)$, the merchant's payoff before paying the platform's fee by $\pi(a,\omega)$, and the platform's direct payoff, before receiving the fee, by $v(a,\omega)$.[7]

By standard results from the information-design literature (e.g., Bergemann and Morris, 2016), the platform's second-period problem can be written as a linear program: without loss of generality, rather than choosing an information structure, the platform chooses a *direct* recommendation mechanism $x : \Omega \to \Delta(A)$, where $x(a|\omega)$ denotes the probability that the merchant is recommended to take action a when the consumer is of type ω. Given database q, the platform chooses a mechanism x and a fee T to solve

$$
\begin{aligned}
V(q) \triangleq \max_{x:\Omega\to\Delta(A),\, T\in\mathbb{R}} \quad & \sum_{a,\omega} v(a,\omega)\, x(a|\omega) q(\omega) + T \\
\text{such that} \quad & \sum_{\omega} \big(\pi(a,\omega) - \pi(a',\omega)\big) x(a|\omega) q(\omega) \geq 0, \ \forall\, a, a' \in A, \qquad \text{(OB)} \\
& \sum_{a,\omega} \pi(a,\omega)\, x(a|\omega) q(\omega) - T \geq 0. \qquad \text{(IR)}
\end{aligned}
$$

A few remarks about the platform's problem are in order. The constraints (OB) are the

[7] The consumer's trading surplus $u(a,\omega)$ already incorporates her optimal response to the merchant's action a. For example, in a price-discrimination setting with unitary demand, a is the price at which the merchant sells the product, ω is the consumer's willingness to pay for such a product, and $u(a,\omega) = \max\{\omega - a, 0\}$.

merchant's *obedience* constraints: conditional on receiving recommendation a, the merchant must be willing to choose a rather than choosing any alternative action a'. These constraints capture the fact that the merchant's action is not contractible. Constraint (IR) is the merchant's participation constraint. Since the platform's objective is increasing in the fee T, this constraint must bind at the optimum. This pins down T and allows the platform to extract the merchant's entire expected payoff. Therefore, the platform's problem can be rewritten as follows:

$$V(q) = \max_x \sum_{a,\omega} \big(v(a,\omega) + \pi(a,\omega)\big) x(a|\omega) q(\omega), \text{ subject to (OB).} \tag{$\mathcal{P}_q$}$$

This is a standard information-design problem, with the distinctive feature that its "prior"—namely, the composition of the database q—is endogenously determined by the competitive data market in the first period.

Solution Concept. We now introduce the equilibrium concept for this competitive data economy.

Definition 1. *A profile (p^*, q^*, x^*) is an **equilibrium** of the competitive economy if*

(a) *Given p^*, q^* solves the platform's problem in the first period, that is,*

$$q^* \in \arg\max_{q \in \mathbb{R}^{\Omega}_{+}} V(q) - \sum_{\omega} p^*(\omega) q(\omega).$$

(b) *Given q^*, x^* solves the platform's second-period problem ($\mathcal{P}_q$).*

(c) *Given p^* and x^*, a type-ω consumer with outside option r sells her record if*

$$r \leq p^*(\omega) + \sum_a x^*(a|\omega) u(a,\omega) \triangleq r^*(\omega).$$

(d) *Data markets clear, that is, $q^*(\omega) = \bar{q}(\omega) F_{\omega}(r^*(\omega))$ for all $\omega \in \Omega$.*

The first two conditions require the platform to behave optimally in the first and second periods, respectively: in the first period, the platform takes data prices as given and anticipates that it will use the acquired database optimally in the second period. The third condition captures the optimality of consumer behavior in the first period: a type-ω consumer sells her record if its price plus the expected trading surplus from participating in the second period is at least as large as her outside option. The last condition requires market clearing: for every type of record, demand equals supply.

The equilibrium has several important properties that will be useful throughout the analysis. First, an equilibrium of the data economy always exists (see Proposition A.4 in the Appendix). Second, in any equilibrium, both the platform and the merchant earn zero net payoff. The platform's profit is zero because it operates in a competitive market: V is homogeneous of degree one, so if the platform earned strictly positive profit at q^*, it could scale up the database and further increase its profit indefinitely, violating market clearing. The merchant earns zero because its payoff is fully extracted through the platform's access fee. Third, the platform's equilibrium database is interior, that is, $0 < q^*(\omega) < \bar{q}(\omega)$ for all ω, because F_ω has full support. Finally, the equilibrium data price for each record type equals the platform's marginal value of that record, i.e., $p^* \in \partial V(q^*)$ (see Lemma A.1 in the appendix).

2.1 Discussion

We briefly pause to discuss the assumptions of our model.

While the model is stylized, our main results extend to a much broader class of environments. First, we can allow for finitely many heterogeneous platforms that compete for the exclusive right to use consumers' data records. Second, each platform can accommodate any finite number of downstream agents, such as multiple competing merchants, advertisers, or bidders. Third, downstream agents may multi-home across platforms, provided their payoffs are additive across platforms and there are no capacity constraints, and they may have bargaining power vis-à-vis the platform. Fourth, these downstream agents and the consumers may interact in any (finite) game, including auctions and many forms of competition—for example, price, quality, and horizontal competition. Finally, each platform may also commit to a data-contingent action that affects downstream payoffs, thereby shaping not only the information available to downstream agents but also the game they play—as when a platform ranks products on a webpage based on a consumer's type. These extensions are discussed in Online Appendix F.[8]

This generality reassures us that the inefficiency we highlight is not an artifact of the stylized baseline model, but a force that can arise in a broad class of environments in which data users act as information intermediaries—which include many platform-mediated markets, such as

[8]The generality discussed in this paragraph is possible because, within this broader class of intermediation problems, the platform's continuation payoff from acquiring a database q, namely $V(q)$, can still be written as the value of a linear program analogous to ($\mathcal{P}_q$).

online marketplaces and advertising-auction platforms.

Despite this generality, three of the assumptions we made deserve discussion.

First and foremost, we assume the platform is a price taker in the data market, so that this market is competitive. This is the central assumption of the paper. Many real-world platforms have, of course, substantial market power. Nevertheless, perfect competition is a natural and important benchmark, as already discussed in the Introduction. Note that we also assumed that the platform has full bargaining power vis-à-vis the merchant. However, this assumption is not essential for our main results (see Appendix F). Its role is simply to shut down downstream distortions that would otherwise arise from imperfect contracting between the platform and the merchant, thereby allowing us to cleanly isolate the data-market distortions that are the focus of the paper.

Second, the concept of a data record bundles two distinct objects: the *exclusive* right to access the consumer and receive information about her. Exclusivity ensures that data are rival goods. We do not allow the consumer to unbundle the access and information—for example, by granting platform access without disclosing her type—although in some settings separating these two dimensions may be important (see, e.g., Ali et al., 2023; Hidir and Vellodi, 2021). Additionally, we assume that a data record fully reveals ω. Imperfectly informative data records would introduce an additional, albeit quite standard, adverse-selection distortion. We discuss this distortion in Appendix E.2.[9]

Third, we assume that the platform is a gatekeeper: consumers and the merchant cannot interact except through the platform. Technically, this assumption ensures that the outside option r is exogenous and thus makes the model tractable. This assumption fits settings, such as some online retail platforms, in which merchants are small and depend on the platform for access to consumers, making the platform a *de facto* gatekeeper. In other settings, however, "offline" interactions matter and can discipline the platform's behavior (see, e.g., Bergemann and Bonatti, 2024, for an analysis of the monopolist problem in this setting).

[9] Since data records bundle access and information and fully reveal ω, consumers and records are in a one-to-one correspondence. Hence our model can be equivalently interpreted as one in which consumers decide whether to join the platform's pool of participants. The price $p(\omega)$ is then the participation payment offered to a type-ω consumer, and the database q describes the composition of the resulting participant pool.

3 The (In)Efficiency of the Data Economy

This section studies whether the first welfare theorem applies to this data economy, that is, whether equilibria maximize social welfare. Section 3.1 introduces the efficiency benchmark, Section 3.2 identifies the data externality at the heart of this economy, and Section 3.3 characterizes when this externality leads to inefficiency and when it does not.

3.1 The Efficiency Benchmark

Fix a database q and a recommendation mechanism x. We denote by $W(q,x)$ the **social welfare** generated by (q,x), defined as the sum of the expected payoffs of the platform, the merchant, and consumers. Formally,

$$W(q,x) \triangleq \sum_{a,\omega} \Big(v(a,\omega) + \pi(a,\omega) + u(a,\omega) \Big) x(a|\omega) q(\omega) + R(q), \tag{1}$$

where $R(q) \triangleq \sum_{\omega} \bar{q}(\omega) \int_{q(\omega)/\bar{q}(\omega)}^{1} F_{\omega}^{-1}(s)\, ds$ is the aggregate outside option of consumers who do not sell their records to the platform.[10] Data prices and the fee charged by the platform to the merchant do not enter the welfare formula because they are simply transfers among market participants, and payoffs are quasi-linear in money.

Conveniently, in equilibrium, social welfare coincides with consumer welfare because, as discussed earlier, both the platform and the merchant must earn zero net payoff.

We call an allocation efficient if its database maximizes social welfare, subject to the platform using that database in a sequentially rational way. To formalize this, let $\mathcal{X}(q)$ denote the set of optimal recommendation mechanisms given database q, i.e., the set of solutions to problem ($\mathcal{P}_q$).

Definition 2. *An allocation (q,x) is **efficient** if it solves*

$$\begin{aligned} \max_{q,x} \quad & W(q,x) \\ \textit{s.t.} \quad & 0 \le q \le \bar{q} \textit{ and } x \in \mathcal{X}(q). \end{aligned}$$

[10] To understand the expression for $R(q)$, note that by parts (c) and (d) of Definition 1, when the platform acquires a mass $q(\omega)$ of type-ω records, the consumers who keep their records are those whose outside option exceeds $F_{\omega}^{-1}\big(\frac{q(\omega)}{\bar{q}(\omega)}\big)$. Note that, without loss of generality, we assume the planner selects the $q(\omega)$-mass of type-ω consumers with the lowest outside option: any other selection rule would weakly reduce social welfare.

In other words, a social planner maximizes social welfare by dictating, for each type ω, which consumers should give their data records to the platform and which should not. The planner is constrained by the fact that it cannot create data records out of thin air and cannot force the platform to use the database in a sequentially irrational way. Subject to platform sequential rationality, the highest welfare the planner can obtain at database q is $\widehat{W}(q) \triangleq \max_{x \in \mathcal{X}(q)} W(q,x)$, which we call the *reduced welfare function*.

Before proceeding, we introduce a *local* notion of efficiency that will be useful later.

Definition 3. *An allocation (q^*, x^*) is **locally efficient** if it maximizes social welfare among all feasible allocations (q,x) whose database q is sufficiently close to q^*. Formally, there is $\delta > 0$ such that $W(q^*, x^*) = \widehat{W}(q^*)$ and $\widehat{W}(q^*) \geq \widehat{W}(q)$ for all $\|q - q^*\| < \delta$.*

An allocation is locally efficient if the planner cannot increase welfare by making small adjustments to it. Clearly, efficiency implies local efficiency. The converse also holds whenever the reduced welfare $\widehat{W}$ is concave in q.

3.2 The Data Externality

We now formalize the central observation of the paper: a consumer's decision to sell her record may impose an externality on other consumers. By selling her record, a consumer changes the *composition* of the platform's database, which may in turn change how the platform uses the data—that is, it may alter the optimal mechanism x. This change affects the trading surplus of other participating consumers, generating a non-pecuniary externality. We will show that consumers internalize the effect of selling their records on their own payoff and, through the competitive data price, on the platform's and merchant's payoff, but not this externality.

To characterize this externality transparently, we first impose an assumption that allows us to use standard derivatives and keep the notation simple. This assumption is not substantive for the results in this subsection.[11]

Definition 4. *A database q is **regular** if $\mathcal{X}(q)$ is a singleton and V is differentiable at q.*

We refer to this as "regularity" because the platform's problem ($\mathcal{P}_q$) is a finite linear program, so V is piecewise linear and differentiable almost everywhere in q. Moreover, generically in

[11] Appendix A.2 drops this assumption and establishes an analogous characterization in terms of directional derivatives.

the payoffs v and π, the platform's problem ($\mathcal{P}_q$) has a unique solution at every interior q, that is, $\mathcal{X}(q)$ is a singleton. Thus, in the unrestricted finite-payoff environment, irregular databases are knife-edge cases; see Appendix D.

We are now ready to identify the source of inefficiency in this data economy. We do so by comparing a consumer's *private benefit* from selling her data with the *social benefit* generated by that sale.

Fix an equilibrium (p^*, q^*, x^*) and suppose that q^* is regular. Consider the marginal consumer who in equilibrium is indifferent between selling and keeping her record. What is her *private benefit* from selling it? From condition (c) in Definition 1, this consumer's indifference condition is

$$p^*(\omega) + U(\omega, x^*) - r^*(\omega) = 0, \tag{2}$$

where $U(\omega, x^*) \triangleq \sum_a x^*(a|\omega)u(a, \omega)$ denotes this consumer's expected trading surplus under the equilibrium mechanism x^*.

What is instead the *social* benefit from having this marginal type-ω consumer sell her record to the platform? Since q^* is regular, there is a locally unique and differentiable selection $q \mapsto x_q \in \mathcal{X}(q)$ with $x_{q^*} = x^*$ (see Lemma A.7). Along this selection, social welfare from equation (1) can be written as

$$W(q, x_q) = V(q) + \sum_{\omega'} q(\omega')U(\omega', x_q) + R(q).$$

The first term is the platform's continuation payoff, which includes the merchant's payoff extracted through the optimal access fee. The second term is the aggregate trading surplus of consumers who join the platform. The third term is the aggregate outside option of consumers who do not sell their records. Differentiating this expression at q^* gives

$$\left.\frac{\partial W(q, x_q)}{\partial q(\omega)}\right|_{q=q^*} = p^*(\omega) + U(\omega, x^*) - r^*(\omega) + \varepsilon_{q^*}(\omega), \tag{3}$$

where

$$\varepsilon_{q^*}(\omega) \triangleq \sum_{a,\omega'} q^*(\omega')u(a, \omega')\left.\frac{\partial x_q(a \mid \omega')}{\partial q(\omega)}\right|_{q=q^*}. \tag{4}$$

Equation (3) uses two equilibrium properties. First, the competitive data price equals the platform's marginal value of the record (since q^* is regular, we can write $p^*(\omega) = \partial V(q^*)/\partial q(\omega)$). Second, the marginal outside option of the type-ω consumer added to the database is $r^*(\omega)$.

More importantly, Equation (4) identifies the externality generated by the sale of a data record. The term $\varepsilon_{q^*}(\omega)$ captures how a marginal increase in the mass of type-ω records in the database changes the platform's mechanism and, through it, the trading surplus of the inframarginal consumers. We refer to this term as the *data externality*.

Comparing (2) and (3), we conclude that consumers internalize the effect of selling their records on their own expected trading surplus and, through the data price $p(\omega)$, on the platform's and merchant's payoffs, but fail to internalize the effect of their participation on the trading surplus of other consumers. This gives our first result.

Proposition 1. *Fix an equilibrium (p^*, q^*, x^*) such that q^* is regular. The equilibrium is locally efficient if and only if there are no data externalities, that is, if and only if $\varepsilon_{q^*}(\omega) = 0$ for all ω. If, in addition, the reduced welfare $\widehat{W}$ is concave in q, then the equilibrium is efficient.*

Therefore, the absence of data externalities is not only necessary but also sufficient for local efficiency. The latter follows from the observation that, at a regular database, the reduced welfare $\widehat{W}$ is locally concave, so first-order conditions certify local optimality. When, in addition, $\widehat{W}$ is concave, efficiency also follows.[12]

3.3 Data Externalities and the Platform's Use of the Database

The previous subsection identified the data externality at the heart of the inefficiency in the economy. We now show that this externality is tightly linked with how the platform uses the database. This externality is present when the equilibrium recommendation mechanism is "sensitive" to the composition of the database; that is, when a small change in q^* alters x^*, thereby reshaping the payoffs of other consumers. Conversely, the externality vanishes when x^* is insensitive to such a change. As we show, this sensitivity is tied to a simple property of the platform's mechanism, which we now introduce.

Definition 5. *A recommendation mechanism $x : \Omega \to \Delta(A)$ is* ***deterministic*** *if $x(a|\omega) \in \{0, 1\}$ for every (a, ω). It is* ***nondeterministic*** *otherwise.*

[12] As discussed in the introduction, Galperti et al. (2024) observe that the marginal value the platform assigns to an ω-record, $\partial V(q)/\partial q(\omega)$, differs from the average payoff generated by records of that type, so that the platform's demand for data may exhibit complementarities. Our setting, however, features a data market that is perfectly competitive. Hence, in equilibrium, each record is priced at its marginal value: $p^*(\omega) = \partial V(q^*)/\partial q(\omega)$ (Lemma A.1). These complementarities are therefore priced in and, hence, do not cause inefficiency.

Notable examples of deterministic mechanisms include *full disclosure*, in which the merchant learns each consumer's type; and *no disclosure*, in which the merchant receives no information about consumers' types. Under a deterministic mechanism, all consumers of the same type receive the same trading surplus ex post. Conversely, under a nondeterministic mechanism, consumers of the same type may experience different outcomes. Our next result shows that such differential treatment typically leads to inefficiency.

Proposition 2. *Generically, every equilibrium with a nondeterministic mechanism x^* is inefficient.*[13]

This result establishes that nondeterministic mechanisms are associated with inefficiency in the data economy. This observation raises concerns about the efficiency of competitive data markets because, as the information design literature has shown, nondeterministic mechanisms emerge as the platform's optimal choice across a wide range of applications, including third-degree price discrimination with a monopolist merchant (Bergemann et al., 2015; Roesler and Szentes, 2017), price competition among multiple merchants (Elliott et al., 2025), consumer-merchant matching (Condorelli and Szentes, 2026), and privacy and consumer profiling by platforms (Strack and Yang, 2024; Fainmesser et al., 2025).

While Proposition 2 does not depend on regularity, its intuition is easiest to see when q^* is regular. Recall that the data externality $\varepsilon_{q^*}(\omega)$ in Equation (4) arises when the equilibrium mechanism is sensitive to the database. A key step in the intuition is that nondeterministic optimal mechanisms are generally sensitive to the database. When x^* is nondeterministic, the platform splits consumers of the same type across different segments. This is typically done to induce certain threshold posterior beliefs for the merchant, leaving him indifferent between two or more actions. In other words, at least one of the merchant's obedience constraints binds at q^*. Hence, if q^* changes marginally, the platform must adjust the recommendation probabilities in order to keep the relevant obedience constraints satisfied. Thus, the optimal mechanism is sensitive to q. By Proposition 1, efficiency then requires these induced changes in the mechanism to have no effect on the aggregate trading surplus of other consumers. This is a strong requirement: the resulting changes in other consumers' surplus must cancel exactly—a restriction on the primitives that can hold only on a measure-zero set.

Next, we give a partial converse of the inefficiency result just discussed.

[13] Genericity is with respect to the Lebesgue measure on the space of primitives $(u, v, \pi, \bar{q})$, with F held fixed.

Proposition 3. *Fix an equilibrium (p^*, q^*, x^*) such that q^* is regular. If x^* is deterministic, then the equilibrium is locally efficient. If, in addition, the reduced welfare $\widehat{W}$ is concave in q, then the equilibrium is efficient.*

The key idea for this result is that, if x^* is deterministic, the optimal mechanism is not sensitive to small perturbations in the database. Thus, the data externality in Equation (4) vanishes, and Proposition 1 implies local efficiency. Intuitively, when the mechanism is deterministic, all participating consumers with the same type receive the same downstream treatment by the merchant. That is, the platform is not fine-tuning the compositions of the segments to make the merchant indifferent between actions. Generally speaking, this means that, when x^* is deterministic, the merchant's incentive constraints are slack at q^*. As a result, a small change in q alters neither the merchant's obedience nor the platform's optimal recommendation for each type. As a result, the same mechanism x^* remains optimal to the platform, and thus is not sensitive to q.

It is worth stressing that, unlike in Proposition 1, the role of regularity in Proposition 3 is not merely expositional. If the platform's value function V were not differentiable at q^*, even a deterministic mechanism might be sensitive to small changes in the database.[14] There is, however, an important case in which this sensitivity cannot arise, and the conclusion of Proposition 3 holds even without regularity.

Corollary 1. *Fix an equilibrium (p^*, q^*, x^*) and suppose that x^* implements full disclosure and is the platform's unique optimal mechanism at q^*. Then the equilibrium is efficient.*[15]

Intuitively, if full disclosure is optimal for the platform at an interior database, it remains optimal at every database. The optimal mechanism is therefore unchanged as the database varies, so the data externality vanishes. Moreover, under full disclosure, the welfare function is concave, and efficiency follows. The corollary also clarifies why the inefficiency we identify is specific to information intermediation. If the platform's objective were aligned with the merchant's payoff—for example, if the platform and the merchant were integrated into the same entity—then full disclosure would be optimal and the data externality would vanish.

[14] See Example E.1 in the Online Appendix. Proposition A.3 in the Appendix extends Proposition 3 to a setting that partially relaxes regularity.

[15] Uniqueness is stronger than needed. This result holds as long as x^* is chosen from $\mathcal{X}(q^*)$ by breaking ties in favor of social welfare.

Hence, inefficiency can arise only when the intermediary has reasons to withhold some of the information contained in the database, as is often the case in platform-mediated markets such as online marketplaces and advertising platforms.

4 An Application: Price Discrimination

So far, we have imposed little structure on the downstream interaction mediated by the platform. Depending on the application, the consumer's type ω, the merchant's action a, and the payoff functions may have different interpretations. In this section, we specialize our model to one such application: the canonical third-degree price-discrimination environment of Bergemann et al. (2015).

In this application, ω is the consumer's willingness to pay for the merchant's product, and a is the price charged by the merchant. A type-ω consumer buys if and only if $a \leq \omega$, so that $u(a, \omega) = \max\{\omega - a, 0\}$ and $\pi(a, \omega) = a\mathbb{1}\{\omega \geq a\}$, where the marginal production cost is assumed to be zero. We assume $\Omega \subset \mathbb{R}_{++}$, so that trade is always efficient and, without loss of generality, let $A = \Omega$. Beyond the access fee charged to the merchant, suppose the platform directly values consumer surplus by assigning it a weight $\gamma \geq 0$: formally, $v(a, \omega) = \gamma\, u(a, \omega)$, so that in problem ($\mathcal{P}_q$) the platform seeks to maximize $\pi(a, \omega) + \gamma\, u(a, \omega)$.[16]

Fix a database q and consider the platform's intermediation problem ($\mathcal{P}_q$). By Bergemann et al. (2015, Theorem 1), the set of feasible pairs of merchant profit and consumer surplus induced by all recommendation mechanisms is the triangle depicted in Figure 1. *Full disclosure*—defined by $x(\omega|\omega) = 1$ for every $\omega \in \Omega$—induces the outcome labeled B: the merchant earns the complete-information monopoly profit, and consumers earn zero trading surplus. *No disclosure*—namely, when $x(a|\omega) = 1$ for some fixed $a \in A$ and every $\omega \in \Omega$—induces the outcome labeled C: the merchant earns the uninformed monopoly profit and consumers capture some of the surplus. Finally, a consumer-optimal mechanism induces the outcome labeled D: the merchant again earns the uninformed monopoly profit, and consumers capture all the additional surplus.

[16] Xu and Yang (2026) offer a dynamic microfoundation for a similar linear payoff specification. They argue that, although the platform profits *directly* from charging merchants, it may value consumer surplus *indirectly*: rents left to consumers today induce them to return in the future, generating additional merchant revenue over time.

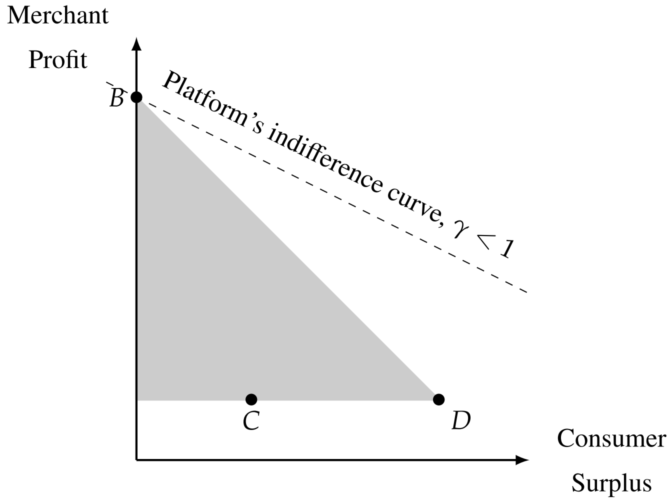

Figure 1: The Bergemann et al. (2015)'s triangle. Each point corresponds to a feasible outcome that the platform can induce by choosing an appropriate recommendation mechanism.

Which outcome the platform finds optimal depends on γ. When $\gamma < 1$, the platform places relatively more weight on extracting the merchant's payoff, its indifference curves are flatter, and full disclosure is (uniquely) optimal. When $\gamma > 1$, the platform places relatively more weight on consumer surplus, its indifference curves are steeper, and a consumer-optimal mechanism is optimal. These two regimes have very different efficiency implications, as the next result shows.

Proposition 4. *Fix $\gamma \geq 0$ and let (p^*, q^*, x^*) be any equilibrium.*

(i) *If $\gamma < 1$, the equilibrium features full disclosure and is efficient.*

(ii) *If $\gamma \geq 1$, the equilibrium is efficient if and only if x^* involves no disclosure.*

In other words, the equilibrium is inefficient whenever the platform uses the database in a nontrivial way, that is, when it neither fully discloses consumers' types nor fully conceals them.

To build intuition, notice that when $\gamma < 1$, full disclosure is the platform's unique optimal mechanism at every database. In this case, the merchant's price for a type-ω consumer is $a = \omega$, independently of the composition of the rest of the database. Hence a marginal change in the mass of one type does not alter how any other types are treated, and the data externalities identified in Section 3.2 are absent. Moreover, since social welfare is concave in q in this case, Proposition 3 implies that all equilibria are efficient.

The case $\gamma \geq 1$ is more subtle. When $\gamma > 1$, the platform finds it uniquely optimal to choose a consumer-optimal mechanism. For $\gamma = 1$, instead, the platform is indifferent among all mechanisms that maximize total surplus. We focus on the $\gamma > 1$ case; the intuition for the $\gamma = 1$ case is similar.

If no-disclosure is optimal for the platform at the database q^*, the merchant must be willing to charge the lowest possible price even absent additional information. For simplicity, suppose that this lowest price is the merchant's unique optimal price. In this case, any small perturbation of q^* leaves the merchant's behavior unchanged, since no-disclosure remains optimal in a neighborhood of q^*. Therefore, the mechanism is insensitive to the database, data externality vanishes, and the equilibrium is efficient.

If instead no-disclosure is suboptimal for the platform, the consumer-optimal outcome cannot be implemented by a deterministic information structure. It requires a nondeterministic policy that partially pools consumers across valuations: some high-valuation consumers must be assigned to the same segment as lower-valuation consumers, so that the merchant is willing to charge them a lower price. The inefficiency is easiest to see by considering a marginal increase in the mass of the lowest consumer type $\underline{\omega}$ in the platform's database. This change increases social welfare because low-valuation consumers can drive down the price for high-valuation consumers. However, the lowest consumer type herself always obtains zero trading surplus, and her participation decision reflects only the market price of her record, which in equilibrium equals the platform's marginal value. This wedge between her private and social values leads to inefficiency. We make this intuition concrete in the following binary-type example.

Example 1. Let $\Omega = \{1, 2\}$, and write $q_i \triangleq q(i)$. If the merchant observes no information, price 1 yields revenue $q_1 + q_2$, while price 2 yields revenue $2q_2$. Thus, when $q_1 < q_2$, the uninformed merchant strictly prefers price 2.

Suppose $\gamma > 1$ and consider an equilibrium with $q_1^* < q_2^*$. Since no disclosure would induce price 2, the consumer-optimal mechanism must partially pool the two types. It assigns all type-1 consumers to price 1, assigns a fraction q_1/q_2 of type-2 consumers to price 1, and assigns the remaining type-2 consumers to price 2. Equivalently, $x_q(1|1) = 1$, $x_q(1|2) = q_1/q_2$, and $x_q(2|2) = 1 - q_1/q_2$. Thus $U(1, x_q) = 0$ and $U(2, x_q) = q_1/q_2$.

The data externalities in Equation (4) are therefore nonzero. Adding type-1 records relaxes the obedience constraint of the price-1 pool and allows more type-2 consumers to receive the low price; formally, $\varepsilon_{q^*}(1) = q_2^* \cdot (1/q_2^*) = 1 > 0$. Adding type-2 records has the opposite effect: it tightens the obedience constraint and forces the platform to reduce the fraction of type-2 consumers assigned to price 1; formally, $\varepsilon_{q^*}(2) = q_2^* \cdot (-q_1^*/(q_2^*)^2) = -q_1^*/q_2^* < 0$. By Proposition 1, the equilibrium is not efficient.

5 Restoring Efficiency in the Data Economy

The inefficiency of the competitive data market raises a natural question: which alternative market arrangements can restore efficiency, and which cannot? This section studies several possibilities. Some build on our baseline model by introducing data taxes, vertically integrating the merchant and the platform, or considering a richer price system. Others depart more substantially from the baseline model by introducing market power in the data market.

5.1 Introducing Market Power

We begin by relaxing the central assumption of the paper that the data market is competitive. We consider two departures from this benchmark: first, a monopsonist platform with market power over consumers; second, a *data union* that gives consumers market power over the platform.

Monopsonist Platform. Suppose that the platform is a *price maker* in the data market, rather than a price taker. The environment is otherwise identical to that of Section 2. More specifically, in the first period, the platform posts data prices p and announces a mechanism x. Given p and x, consumers then choose whether to keep or sell their data records. Because the platform lacks commitment power, however, not every announced mechanism is credible—once the database has been acquired, the platform will carry out the announced mechanism only if it is sequentially rational to do so. Formally, the monopsonist platform solves

$$\begin{aligned} \max_{p,x} \quad & V(q_{p,x}) - \sum_{\omega} p(\omega) q_{p,x}(\omega) \\ \text{s.t.} \quad & q_{p,x}(\omega) = \bar{q}(\omega) F_{\omega}\big(p(\omega) + U(\omega, x)\big) && \text{(5a)} \\ & x \in \mathcal{X}(q_{p,x}). && \text{(5b)} \end{aligned}$$

In other words, the monopsonist chooses prices and the mechanism anticipating how they jointly shape consumer participation and its own continuation payoff. A *monopsony equilibrium* is a profile (p^M, q^M, x^M) such that $q^M = q_{p^M, x^M}$ and (p^M, x^M) solves the monopsonist's problem.

Remark 1. *Every monopsony equilibrium is inefficient.*

This inefficiency is not driven by the data externality: because the platform can set prices, it internalizes the effect of the database composition on consumers' trading surplus. Instead, the inefficiency reflects a standard monopsony distortion: acquiring an additional type-ω record forces the platform to raise $p(\omega)$, and this higher price must be paid not only to the marginal consumer but to all inframarginal ones. This induces the monopsonist platform to mark down data prices below their marginal value and underacquire data records—a standard monopsony distortion that does not depend on the fact that the good being traded is data.

Data Union. Next, we consider a *data union*—an institution that manages consumers' data records on their behalf, sells the resulting database to the platform, and redistributes the proceeds among its members. Versions of this idea have been proposed by Posner and Weyl (2018) and Bergemann et al. (2023). As we show, a suitably designed data union restores efficiency.

The data union sets prices p and collects data records from the consumers. It then sells the database to the platform with a take-it-or-leave-it offer. The platform then uses the data as described in Section 2 (with potential ties broken in favor of the data union). Specifically, the data-union problem is

$$\begin{aligned} \max_{p,x} \quad & \sum_{\omega} q_{p,x}(\omega)\big(p(\omega) + U(\omega, x)\big) - \sum_{\omega} \bar{q}(\omega) \int_0^{q_{p,x}(\omega)/\bar{q}(\omega)} F_{\omega}^{-1}(s)\, ds \\ \text{s.t.} \quad & \text{(5a), (5b), and } \sum_{\omega} q_{p,x}(\omega) p(\omega) \leq V(q_{p,x}) \end{aligned}$$

In words, the data union's objective is to maximize the welfare of its members: their payoff from participating, $p(\omega) + U(\omega, x)$, net of the outside options they forgo when their data are sold to the platform, as captured by the last term in the objective. The last constraint requires the data union to run no deficit: the total transfer paid to its members cannot exceed the revenue it raises from selling the database to the platform, which equals the platform's continuation surplus $V(q_{p,x})$.

A *data-union equilibrium* is a profile (p^U, q^U, x^U) such that $q^U = q_{p^U, x^U}$ and (p^U, x^U) solves the data-union problem.

Remark 2. *An allocation is efficient if and only if it is induced by a data-union equilibrium.*

The data union thus restores efficiency in the data economy. Like a monopsonist platform, the union internalizes the data externality. Unlike a monopsonist platform, however, it does not distort data acquisition to reduce payments to inframarginal consumers, since those payments are redistributed to the union's own members. The monopsony distortion therefore disappears, and the union's objective coincides with social welfare up to an additive constant.

5.2 Other Remedies

Contractible Actions and Vertical Integration. Imagine that the platform—rather than acting only as an information intermediary—could write an enforceable contract specifying which action the merchant must take for each record in its database. In this case, the platform can discipline the merchant's actions directly, rather than designing an information structure that makes the desired behavior incentive compatible. The obedience constraints therefore drop from the platform's problem, and the data externality vanishes (see Remark C.1 in the Appendix).[17] The same logic also clarifies what would happen under vertical integration: if the platform and the merchant were a single entity, the platform could choose the downstream action for each record, again eliminating the data externality.

Data Taxes. A classic solution to an externality is to introduce taxes (or subsidies). The tax levied on the sale of an ω-record equals the negative of the externality that type generates at the efficient allocation, $\tau^*(\omega) = -\varepsilon_{q^*}(\omega)$: types that create a positive externality on others are subsidized, while those that impose a negative one are taxed. By making each consumer internalize the effect of her participation on the trading surplus of other consumers, the tax restores the alignment between private and social incentives (see Remark C.2 in the Appendix). We also show that the scheme requires no outside funding: the government runs a balanced budget. Importantly, however, implementing these taxes requires the government to know the type-specific data externalities $\varepsilon_{q^*}(\omega)$, which is likely impractical.

Lindahl Pricing. Finally, we enrich the price system itself. We let the price of a record depend not only on its type ω but also on the use to which it is put; that is, we open a separate mar-

[17] Vertical integration restores efficiency relative to a different benchmark: the platform chooses the mechanism optimally given that the actions are contractible.

ket for each action–type pair (a, ω). These Lindahl prices lead participants to internalize the externalities associated with each distinct use of the data. The resulting economy implements an even stronger benchmark, in which the planner chooses both the database and the recommendation mechanism subject only to merchant obedience, dropping the requirement that the mechanism be sequentially optimal for the platform. Lindahl pricing thus corrects not only the data externality but also the distortion arising from the platform's inability to commit, in the first period, to its second-period mechanism (see Remark C.3 in the Appendix).

6 Conclusions

This paper studied competitive markets for personal data. The central lesson is that, when data users are information intermediaries—as is often the case in platform-mediated markets, such as online marketplaces and advertising-auction platforms—competition in the data market does not, by itself, guarantee efficiency. Indeed, when a consumer sells her record, she changes the composition of the intermediary's database; this may change what information the intermediary provides to merchants and, therefore, how other consumers are treated—even if the sold record is entirely uninformative about these consumers. Competitive prices do not capture these effects.

The main results of this paper characterize when this externality should be expected to arise and what economic force generates it. In particular, we find that, when the intermediary's equilibrium information policy is nondeterministic—meaning that consumers of the same type may experience different outcomes—the equilibrium is generically inefficient. We argued that nondeterministic mechanisms are far from being a knife-edge possibility. Rather, as the information-design literature has shown, they arise naturally as optimal mechanisms in a broad class of applications. From this perspective, our results raise concerns about the efficiency of competitive data markets in a wide range of environments.

We established these results in a rather stylized environment with one representative platform, one merchant, and full surplus extraction by the platform in the product market. These assumptions are mainly expositional. As discussed in Section 2.1, our main results extend to a broader class of environments, including finitely many heterogeneous platforms and multiple merchants, who may multi-home and retain some bargaining power vis-à-vis the platforms.

Our analysis leaves several open questions that we see as natural next steps. Some concern institutions. The remedies discussed in the previous section often require regulators to have substantial information, which may be unrealistic, or may impose burdens on consumers and firms, which may undermine their practical appeal. Understanding whether simpler policies can approximate the efficiency gains of these solutions is an important open question. Others concern scope. We focused on *personal data*—such as age, gender, and income—that are informative about consumer preferences, treating each consumer's record as given and her only decision as whether to sell it. Other forms of data, however, are not passively held but produced through costly effort. Training data for AI systems—such as writing an essay or driving a car—is a leading example. How data markets would work for this kind of data remains an open question.

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Appendix

A Proofs for Section 3

A.1 Preliminaries

Throughout the appendix we write

$$r_q(\omega) \triangleq F_\omega^{-1}(q(\omega)/\bar{q}(\omega)) \qquad \text{and} \qquad U(\omega, x) \triangleq \sum_a x(a|\omega)u(a,\omega)$$

for the marginal type-ω seller's outside option at q and for type ω's expected trading surplus under x. The notation $r_q(\omega)$ is used only when $0 < q(\omega) < \bar{q}(\omega)$. We also use repeatedly that $\mathcal{X}(q)$ depends on q and (v, π) but not on u.

Lemma A.1 (Equilibrium characterization)**.** *A profile (p^*, q^*, x^*) is an equilibrium if and only if $p^* \in \partial V(q^*)$, $x^* \in \mathcal{X}(q^*)$, and $q^*(\omega) = \bar{q}(\omega)F_\omega(p^*(\omega) + U(\omega, x^*))$ for all ω. Every equilibrium satisfies $0 < q^*(\omega) < \bar{q}(\omega)$.*

Proof. Since V is concave, q^* solves the platform's first-period problem $\max_q V(q) - p^* \cdot q$ if and only if $p^* \in \partial V(q^*)$, where $\partial V(q^*)$ denotes the superdifferential of the concave function V at q^*. The second-period optimality condition in Definition 1 is exactly $x^* \in \mathcal{X}(q^*)$. Finally, consumer optimality says that a type-ω consumer sells if and only if $r \leq p^*(\omega) + U(\omega, x^*)$; market clearing is therefore equivalent to

$$q^*(\omega) = \bar{q}(\omega)F_\omega(p^*(\omega) + U(\omega, x^*)) \qquad \forall\omega.$$

Because A and Ω are finite, the cutoff $p^*(\omega) + U(\omega, x^*)$ is finite. Since F_ω has a strictly positive density on $\mathbb{R}$, it has full support and $F_\omega(r) \in (0, 1)$ for every finite r. Thus $0 < q^*(\omega) < \bar{q}(\omega)$ for every ω. ◻

Lemma A.2 (Outside-option term)**.** *R is concave and differentiable on $\prod_\omega(0, \bar{q}(\omega))$, with $\partial R(q)/\partial q(\omega) = -r_q(\omega)$.*

Proof. For each ω, define

$$R_\omega(z) \triangleq \bar{q}(\omega) \int_{z/\bar{q}(\omega)}^{1} F_\omega^{-1}(s)\, ds \qquad \text{for } 0 < z < \bar{q}(\omega).$$

Then $R(q) = \sum_\omega R_\omega(q(\omega))$. By the fundamental theorem of calculus,

$$R'_\omega(z) = -F_\omega^{-1}(z/\bar{q}(\omega)) .$$

Since F_ω^{-1} is increasing, R'_ω is decreasing, so each R_ω is concave. Therefore R is concave and differentiable on $\prod_\omega(0, \bar{q}(\omega))$, with the stated derivative. $\square$

Lemma A.3 (Homogeneity)**.** *V is homogeneous of degree one; $\mathcal{X}(\lambda q) = \mathcal{X}(q)$ for $\lambda > 0$; and $p \in \partial V(q)$ implies $p \cdot q = V(q)$.*

Proof. Fix $\lambda > 0$. Multiplying the database q by λ multiplies the objective in ($\mathcal{P}_q$) and every obedience constraint by λ, while leaving the set of feasible mechanisms unchanged. Thus $\mathcal{X}(\lambda q) = \mathcal{X}(q)$ and $V(\lambda q) = \lambda V(q)$.

Let $p \in \partial V(q)$. Since V is concave, $V(q') \leq V(q) + p \cdot (q' - q)$ for every $q' \geq 0$. Taking $q' = 0$ and using $V(0) = 0$ gives $p \cdot q \leq V(q)$. Taking $q' = \lambda q$ gives

$$\lambda V(q) \leq V(q) + p \cdot (\lambda q - q) = V(q) + (\lambda - 1) p \cdot q.$$

For $\lambda > 1$, this implies $p \cdot q \geq V(q)$. Combining the two inequalities yields $p \cdot q = V(q)$. $\square$

A.2 Characterization of Local Efficiency and Proof of Proposition 1

We say that an equilibrium (p^*, q^*, x^*) is *planner-preferred* if x^* is planner-preferred at q^*, i.e., $W(q^*, x^*) = \widehat{W}(q^*)$. We write $V'(q; d)$ and $\widehat{W}'(q; d)$ for right directional derivatives in direction d.

Lemma A.4 (Regularity of reduced welfare)**.** *$\widehat{W}$ is directionally differentiable on $\prod_\omega(0, \bar{q}(\omega))$, and if $\widehat{W}'(q; d) \leq 0$ for every d, then q is a local maximizer of $\widehat{W}$.*

The proof is a finite-linear-programming argument, deferred to Online Appendix D.

Proposition A.1 (Directional criterion)**.** *A planner-preferred equilibrium (p^*, q^*, x^*) is locally efficient if and only if $\widehat{W}'(q^*; d) \leq 0$ for every $d \in \mathbb{R}^\Omega$.*

Proof. Suppose first that (p^*, q^*, x^*) is locally efficient. Then x^* must be planner-preferred at q^*: otherwise, the planner could keep the same database q^* and choose a mechanism in $\mathcal{X}(q^*)$

that yields strictly higher welfare. Moreover, for every q sufficiently close to q^* and every $x \in \mathcal{X}(q)$,

$$W(q, x) \le W(q^*, x^*) = \widehat{W}(q^*).$$

Taking the maximum over $x \in \mathcal{X}(q)$ gives $\widehat{W}(q) \le \widehat{W}(q^*)$ for all nearby q. Thus q^* is a local maximizer of $\widehat{W}$. Since $\widehat{W}$ is directionally differentiable by Lemma A.4, $\widehat{W}'(q^*; d) \le 0$ for every direction d.

Conversely, suppose that x^* is planner-preferred at q^* and $\widehat{W}'(q^*; d) \le 0$ for every d. Lemma A.4 implies that q^* is a local maximizer of $\widehat{W}$. Therefore, for every feasible (q, x) with q sufficiently close to q^*,

$$W(q, x) \le \widehat{W}(q) \le \widehat{W}(q^*) = W(q^*, x^*).$$

This is exactly local efficiency. $\square$

To evaluate $\widehat{W}'$ we use the following decomposition.

Lemma A.5 (Marginal welfare along an optimal path)**.** *Let (p^*, q^*, x^*) be an equilibrium, $d \in \mathbb{R}^\Omega$, and $t \mapsto x_t$ a right-differentiable path with $x_0 = x^*$ and $x_t \in \mathcal{X}(q^* + td)$ for small $t \ge 0$. Let $\dot{x}_0(a|\omega) \triangleq \frac{d}{dt} x_t(a|\omega)\Big|_{0+}$ denote the right derivative of the path at zero. Then*

$$\frac{d}{dt} W(q^* + td, x_t)\Big|_{0+} = E(d) - \left[p^* \cdot d - V'(q^*; d)\right], \qquad E(d) \triangleq \sum_{a,\omega} q^*(\omega) u(a, \omega) \dot{x}_0(a|\omega),$$

and the bracket is nonnegative. If x_t is planner-preferred for small $t \ge 0$, the left side is $\widehat{W}'(q^*; d)$.

Proof. Let $q_t = q^* + td$. Since $x_t \in \mathcal{X}(q_t)$, the platform value at q_t is attained by x_t, so welfare along the path is

$$W(q_t, x_t) = V(q_t) + \sum_{\omega} q_t(\omega) U(\omega, x_t) + R(q_t).$$

The right derivative of the first term at $t = 0$ is $V'(q^*; d)$. For the consumer-surplus term, write

$$\sum_{\omega} q_t(\omega) U(\omega, x_t) = \sum_{a,\omega} q_t(\omega) x_t(a|\omega) u(a, \omega).$$

Right-differentiability of x_t gives derivative

$$\sum_{\omega} d(\omega) U(\omega, x^*) + \sum_{a,\omega} q^*(\omega) u(a, \omega) \dot{x}_0(a|\omega) = \sum_{\omega} d(\omega) U(\omega, x^*) + E(d).$$

By Lemma A.2, the derivative of $R(q_t)$ at zero is $-\sum_\omega d(\omega) r_{q^*}(\omega)$. Combining these derivatives yields

$$\frac{d}{dt} W(q_t, x_t)\Big|_{0+} = V'(q^*; d) + \sum_\omega d(\omega)\big(U(\omega, x^*) - r_{q^*}(\omega)\big) + E(d).$$

By Lemma A.1, market clearing implies $r_{q^*}(\omega) = p^*(\omega) + U(\omega, x^*)$. Therefore the middle sum is $-p^* \cdot d$, giving the displayed identity.

It remains to check the sign of the bracket. Since $p^* \in \partial V(q^*)$ and V is concave, $V(q^* + td) \le V(q^*) + tp^* \cdot d$ for all small $t > 0$. Dividing by t and taking the right limit gives $V'(q^*; d) \le p^* \cdot d$. Hence $p^* \cdot d - V'(q^*; d) \ge 0$. Finally, if x_t is planner-preferred, then $W(q_t, x_t) = \widehat{W}(q_t)$ for small $t \ge 0$ and $W(q^*, x^*) = \widehat{W}(q^*)$, so the derivative on the left is $\widehat{W}'(q^*; d)$. □

Lemma A.6 (Differentiable selections)**.** *If $q \mapsto x_q \in \mathcal{X}(q)$ is differentiable near q^*, then V is differentiable at q^*; if in addition (p^*, q^*, x^*) is an equilibrium with $x_{q^*} = x^*$, then $p^* = \nabla V(q^*)$.*

Proof. For every q in the neighborhood, x_q solves the platform problem at q. Hence

$$V(q) = \sum_{a,\omega} (v + \pi)(a, \omega) x_q(a|\omega) q(\omega)$$

on that neighborhood. The right-hand side is differentiable in q, because $q \mapsto x_q$ is differentiable. Thus V is differentiable at q^*. If (p^*, q^*, x^*) is an equilibrium with $x_{q^*} = x^*$, Lemma A.1 gives $p^* \in \partial V(q^*)$. Since V is differentiable at q^*, its superdifferential is the singleton $\{\nabla V(q^*)\}$, so $p^* = \nabla V(q^*)$. □

Proposition A.2 (Smooth criterion)**.** *Let (p^*, q^*, x^*) be a planner-preferred equilibrium admitting a differentiable planner-preferred selection $q \mapsto x_q$ with $x_{q^*} = x^*$. Then it is locally efficient iff $\varepsilon_{q^*}(\omega) = 0$ for all ω; if moreover $\widehat{W}$ is concave, it is efficient.*

Proof. By Lemma A.6, $p^* = \nabla V(q^*)$. Hence, for every d, the directional derivative $V'(q^*; d)$ equals $p^* \cdot d$, so the bracket in Lemma A.5 is zero. Apply that lemma to the path $q_t = q^* + td$ and $x_t = x_{q_t}$. Since the selection is planner-preferred, the derivative is $\widehat{W}'(q^*; d)$. Since x_q is differentiable,

$$\dot{x}_0(a|\omega) = \sum_{\omega'} \frac{\partial x_q(a|\omega)}{\partial q(\omega')}\Big|_{q=q^*} d(\omega').$$

Substituting into the definition of $E(d)$ and rearranging the finite sums gives

$$\widehat{W}'(q^*;d) = \sum_{\omega'} d(\omega') \sum_{a,\omega} q^*(\omega) u(a,\omega) \frac{\partial x_q(a|\omega)}{\partial q(\omega')}\bigg|_{q=q^*} = \sum_{\omega'} d(\omega') \varepsilon_{q^*}(\omega'),$$

where the last equality is the definition of the data externality in (4). By Proposition A.1, local efficiency is equivalent to $\widehat{W}'(q^*;d) \le 0$ for every d. Since q^* is interior, both d and $-d$ are feasible local directions. Therefore the linear form $d \mapsto \sum_\omega d(\omega)\varepsilon_{q^*}(\omega)$ is nonpositive in every direction if and only if $\varepsilon_{q^*}(\omega) = 0$ for every ω.

If $\widehat{W}$ is concave, local maximality along the planner-preferred selection implies global maximality. Since the selection is planner-preferred, this is equivalent to efficiency. □

Lemma A.7 (Regular databases have differentiable selections)**.** *If q^* is regular, then on a neighborhood of q^* the platform's optimal mechanism is unique and depends differentiably on q.*

The proof is a local linear-programming argument, deferred to Online Appendix D.

Proof of Proposition 1. At a regular q^*, $\mathcal{X}(q^*)$ is a singleton. Thus the equilibrium mechanism x^* is planner-preferred at q^*. Lemma A.7 gives a differentiable selection $q \mapsto x_q \in \mathcal{X}(q)$ with $x_{q^*} = x^*$; because $\mathcal{X}(q)$ is a singleton in a neighborhood of q^*, this selection is also planner-preferred. Proposition A.2 therefore gives the equivalence between local efficiency and $\varepsilon_{q^*} \equiv 0$. If $\widehat{W}$ is concave, the final claim follows from the last part of Proposition A.2. □

A.3 Proof of Proposition 2

Call platform payoffs (v,π) *regular* if ($\mathcal{P}_q$) has a unique optimal joint allocation $\chi(a,\omega) = x(a|\omega)q(\omega)$ at every interior database; this holds for Lebesgue-a.e. (v,π) (Lemma D.3). The proof of Proposition 2 has two parts. First, scaling a recommendation segment gives a simple necessary condition for local efficiency. Second, for a nondeterministic mechanism this condition is a nontrivial linear restriction on consumer payoffs, and the market-clearing map preserves its nullity.

Lemma A.8 (Segment scaling)**.** *Fix an equilibrium (p^*,q^*,x^*) and an action a used with positive mass; write $q^a(\omega) \triangleq x^*(a|\omega)q^*(\omega)$. For all small t of either sign there is $x_t \in \mathcal{X}(q^* + tq^a)$ with $x_0 = x^*$, and*

$$\frac{d}{dt} W(q^* + tq^a, x_t)\bigg|_{t=0} = \sum_\omega q^a(\omega)\big(u(a,\omega) - U(\omega,x^*)\big). \tag{A.1}$$

Local efficiency requires this sum to vanish.

Proof. Let $\chi^*(b,\omega) = x^*(b|\omega)q^*(\omega)$ be the joint allocation induced by the equilibrium mechanism. For t close to zero, define

$$\chi_t(b,\omega) = \begin{cases} (1+t)\chi^*(a,\omega), & b = a, \\ \chi^*(b,\omega), & b \neq a. \end{cases}$$

For $|t|$ small and $1+t > 0$, χ_t is nonnegative. Its database is $q_t = q^* + tq^a$. Obedience is preserved: the obedience constraints associated with recommendation a are multiplied by $1+t$, while all other obedience constraints are unchanged. Thus χ_t is feasible at q_t.

Let $c(a,\omega) = (v+\pi)(a,\omega)$ and let $\Phi_a = \sum_\omega c(a,\omega)q^a(\omega)$. Feasibility of χ_t gives

$$V(q_t) \geq V(q^*) + t\Phi_a.$$

Since $p^* \in \partial V(q^*)$, concavity gives

$$V(q_t) \leq V(q^*) + tp^* \cdot q^a.$$

These two inequalities hold for both positive and negative small t. For $t > 0$ they imply $\Phi_a \leq p^* \cdot q^a$; for $t < 0$ they imply $\Phi_a \geq p^* \cdot q^a$. Hence $\Phi_a = p^* \cdot q^a$, and the upper and lower bounds coincide. Therefore $V(q_t) = V(q^*) + t\Phi_a$, so χ_t is optimal at q_t. Let x_t be the mechanism induced by χ_t and q_t; then $x_t \in \mathcal{X}(q_t)$ and $x_0 = x^*$.

The welfare derivative along this path is the derivative of the platform value, plus the derivative of consumer surplus, plus the derivative of R. The platform-value derivative is $\Phi_a = p^* \cdot q^a$. The consumer-surplus derivative is $\sum_\omega q^a(\omega)u(a,\omega)$, because the only mass being varied is the mass assigned to recommendation a. By Lemma A.2, the derivative of $R(q_t)$ at zero is $-\sum_\omega q^a(\omega)r_{q^*}(\omega)$. Using market clearing, $r_{q^*}(\omega) = p^*(\omega) + U(\omega, x^*)$. Thus

$$\frac{d}{dt}W(q_t, x_t)\Big|_{t=0} = p^* \cdot q^a + \sum_\omega q^a(\omega)u(a,\omega) - \sum_\omega q^a(\omega)\big(p^*(\omega) + U(\omega,x^*)\big),$$

which simplifies to Equation (A.1). Since the path is feasible for both signs of t, local efficiency requires the derivative to be zero. □

Lemma A.9 (Nullity of efficient split equilibria)**.** *Fix F and regular platform payoffs (v,π). Then the set of $(u,\bar{q})$ for which there exists a locally efficient equilibrium with a nondeterministic mechanism has Lebesgue measure zero.*

The proof is measure-theoretic and is deferred to Online Appendix D.

Proof of Proposition 2. By Lemma D.3, for Lebesgue-a.e. platform payoff primitives (v, π) the platform payoffs are regular in the sense used above. Fix such a regular pair (v, π). Lemma A.9 implies that, holding (v, π, F) fixed, the set of $(u, \bar{q})$ for which there exists a locally efficient nondeterministic equilibrium is Lebesgue null. By Fubini's theorem, the set of primitive profiles $(u, v, \pi, \bar{q})$ admitting a locally efficient nondeterministic equilibrium is Lebesgue null, holding F fixed. Since efficiency implies local efficiency, for generic primitives every equilibrium with a nondeterministic mechanism is inefficient. □

A.4 Proof of Proposition 3

Proposition A.3 (Deterministic mechanisms under a differentiable selection)**.** *Let (p^*, q^*, x^*) be a planner-preferred equilibrium with a differentiable planner-preferred selection $q \mapsto x_q$, $x_{q^*} = x^*$. If x^* is deterministic, the equilibrium is locally efficient; if moreover $\widehat{W}$ is concave, it is efficient.*

Proof. Fix (a, ω). The coordinate map $q \mapsto x_q(a|\omega)$ is differentiable at the interior point q^* and takes values in $[0, 1]$ in a neighborhood of q^*. Since x^* is deterministic, $x_{q^*}(a|\omega)$ is either 0 or 1. If the derivative of this coordinate in some direction d were positive at a point where the value is 1, then for small $t > 0$ the value at $q^* + td$ would exceed 1; if it were negative, then for small $t > 0$ the value at $q^* - td$ would exceed 1. Both are impossible. The same argument at the boundary value 0 shows that every directional derivative is zero. Hence $D_q x_q(a|\omega)|_{q=q^*} = 0$ for every (a, ω).

It follows from (4) that $\varepsilon_{q^*}(\omega) = 0$ for every ω. Proposition A.2 gives local efficiency, and its concavity part gives efficiency when $\widehat{W}$ is concave. □

Proof of Proposition 3. At a regular database, the equilibrium mechanism is the unique element of $\mathcal{X}(q^*)$, hence planner-preferred. Lemma A.7 supplies a differentiable local selection from $\mathcal{X}$ through x^*. Proposition A.3 applies. □

Proof of Corollary 1. We prove a stronger claim: if a planner-preferred equilibrium (p^*, q^*, x^*) has x^* fully revealing, it is efficient.

For each type ω, define

$$\bar{v}(\omega) \triangleq \sum_a x^*(a|\omega)(v+\pi)(a,\omega), \qquad \bar{w}(\omega) \triangleq \bar{v}(\omega) + U(\omega, x^*).$$

Full revelation means that every recommendation used by type ω induces the degenerate posterior on ω. Therefore, for every used recommendation b of type ω, obedience implies $\pi(b,\omega) \geq \pi(a,\omega)$ for every action a; the same type-separating mechanism is feasible at every database.

We first show that no obedient posterior-action pair can improve on the typewise platform payoff vector. Fix a posterior $\mu \in \Delta(\Omega)$ and an action a that is obedient at μ. Since q^* is interior, choose $\eta > 0$ small enough that $\eta\mu(\omega) \leq q^*(\omega)$ for every ω. Starting from the fully revealing implementation at q^*, remove mass $\eta\mu(\omega)$ from type ω, proportionally from the recommendations used by that type, and place the removed mass in a new signal recommending a. The old signals remain degenerate and obedient, and the new signal is obedient by assumption. Platform optimality of x^* at q^* therefore implies

$$\sum_\omega \mu(\omega)\big((v+\pi)(a,\omega) - \bar{v}(\omega)\big) \leq 0. \tag{A.2}$$

If equality holds, the perturbed mechanism is also platform-optimal at q^*. Since x^* is planner-preferred among platform-optimal mechanisms at q^*, the same perturbation cannot raise total continuation welfare, and thus

$$\sum_\omega \mu(\omega)\big((v+\pi+u)(a,\omega) - \bar{w}(\omega)\big) \leq 0. \tag{A.3}$$

Now fix any database q and any obedient mechanism y. For each action a used by y, let $\alpha_a = \sum_\omega y(a|\omega)q(\omega)$ and, when $\alpha_a > 0$, let $\mu_a(\omega) = y(a|\omega)q(\omega)/\alpha_a$. Obedience of y implies that action a is obedient at posterior μ_a. Applying (A.2) to each used a and multiplying by α_a gives

$$\sum_{a,\omega}(v+\pi)(a,\omega)y(a|\omega)q(\omega) \leq \sum_\omega \bar{v}(\omega)q(\omega).$$

The fully revealing mechanism attains the right-hand side, so it is platform-optimal at every database. If $y \in \mathcal{X}(q)$, then equality must hold in the preceding display for every used recommendation with positive mass; applying (A.3) to each such recommendation gives

$$W(q,y) \leq \sum_\omega \bar{w}(\omega)q(\omega) + R(q).$$

Full revelation attains equality, so it is planner-preferred at every database and

$$\widehat{W}(q) = \sum_{\omega} \bar{w}(\omega) q(\omega) + R(q).$$

This function is concave. At the equilibrium database, Lemma A.1 and the linearity of the platform value under full revelation give $p^*(\omega) = \bar{v}(\omega)$. Market clearing gives

$$r_{q^*}(\omega) = p^*(\omega) + U(\omega, x^*) = \bar{w}(\omega).$$

By Lemma A.2, these are exactly the first-order conditions for maximizing the concave function $\widehat{W}$. Thus q^* is a global maximizer of $\widehat{W}$, and the equilibrium is efficient. □

A.5 Equilibrium Existence

Proposition A.4. *An equilibrium of the competitive data economy exists.*

Proof. By Lemma D.1, we can write $V(q) = \min_{1 \le j \le J} a_j \cdot q$ on $\mathbb{R}_+^{\Omega}$. For each q, let

$$I(q) = \{j : V(q) = a_j \cdot q\}, \qquad \Sigma(q) = \mathrm{co}\{a_j : j \in I(q)\}.$$

For a polyhedral concave function written as a finite minimum of linear functions, $\Sigma(q) = \partial V(q)$ for an interior q.

Let $M = \max_j \|a_j\|_\infty$, let $P = [-M, M]^{\Omega}$, let $Q = \prod_\omega [0, \bar{q}(\omega)]$, and let $\mathbb{X} = \prod_\omega \Delta(A)$. Define

$$\Gamma(p, x)(\omega) = \bar{q}(\omega) F_\omega(p(\omega) + U(\omega, x)).$$

Then $\Gamma : P \times \mathbb{X} \to Q$ is continuous. Define the correspondence

$$\mathcal{F}(p, q, x) = \Sigma(q) \times \{\Gamma(p, x)\} \times \mathcal{X}(q)$$

from $P \times Q \times \mathbb{X}$ into itself.

The domain is nonempty, compact, and convex. The set $\Sigma(q)$ is nonempty, compact, and convex. The set $\mathcal{X}(q)$ is nonempty because the platform's second-period feasible set is nonempty, and it is compact and convex because it is the set of maximizers of a linear objective over a compact convex set. Thus $\mathcal{F}$ has nonempty, compact, convex values.

We prove that $\mathcal{F}$ has closed graph. Let $(p_n, q_n, x_n) \to (p, q, x)$ and $(\tilde{p}_n, \tilde{q}_n, \tilde{x}_n) \to (\tilde{p}, \tilde{q}, \tilde{x})$ with $(\tilde{p}_n, \tilde{q}_n, \tilde{x}_n) \in \mathcal{F}(p_n, q_n, x_n)$. Continuity of Γ gives $\tilde{q} = \Gamma(p, x)$. Since $\tilde{p}_n \in \Sigma(q_n)$, write $\tilde{p}_n = \sum_j \lambda_j^n a_j$, where $\lambda_j^n \geq 0$, $\sum_j \lambda_j^n = 1$, and $\lambda_j^n = 0$ whenever $j \notin I(q_n)$. Passing to a subsequence, suppose $\lambda_j^n \to \lambda_j$ for every j. Then $\tilde{p} = \sum_j \lambda_j a_j$. If $\lambda_j > 0$, then $j \in I(q_n)$ for all large n, so by continuity of V we have $V(q) = a_j \cdot q$. Hence $j \in I(q)$, and $\tilde{p} \in \Sigma(q)$.

Finally, $\tilde{x}_n \in \mathcal{X}(q_n)$. The obedience constraints pass to the limit, so $\tilde{x}$ is feasible at q. Optimality also passes to the limit because

$$\sum_{a,\omega} (v + \pi)(a, \omega) \tilde{x}_n(a|\omega) q_n(\omega) = V(q_n)$$

for every n, and both sides are continuous in the limit. Hence $\tilde{x} \in \mathcal{X}(q)$. Therefore the graph of $\mathcal{F}$ is closed.

Kakutani's theorem gives a fixed point (p^*, q^*, x^*). Because p^* and $U(\omega, x^*)$ are finite and F_ω has full support, $q^*(\omega) \in (0, \bar{q}(\omega))$ for every ω. Hence $\Sigma(q^*) = \partial V(q^*)$. At the fixed point, $p^* \in \partial V(q^*)$, $x^* \in \mathcal{X}(q^*)$, and $q^*(\omega) = \bar{q}(\omega) F_\omega(p^*(\omega) + U(\omega, x^*))$ for every ω. Lemma A.1 then implies that (p^*, q^*, x^*) is an equilibrium. □

B Proofs for Section 4

We prove Proposition 4 using the feasible-surplus characterization of Bergemann et al. (2015). Let

$$w(q) \triangleq \sum_\omega \omega q(\omega)$$

be the total gains from trade under full participation, and let $\pi_m(q)$ be the merchant's uninformed monopoly profit. By Bergemann et al. (2015, Theorem 1), the feasible pairs of merchant profit Π and aggregate consumer surplus S are exactly

$$\pi_m(q) \leq \Pi \leq w(q), \qquad 0 \leq S \leq w(q) - \Pi.$$

The platform maximizes $\Pi + \gamma S$, while continuation welfare equals $\Pi + (1 + \gamma) S$.

Proof of Proposition 4. (i) $\gamma < 1$. The platform objective on the BBM triangle is $\Pi + \gamma S$. Since $S \leq w(q) - \Pi$, we have

$$\Pi + \gamma S \leq \Pi + \gamma(w(q) - \Pi) = \gamma w(q) + (1 - \gamma)\Pi.$$

When $\gamma < 1$, the right-hand side is maximized by the largest feasible Π, namely $\Pi = w(q)$. This is the full-disclosure point, with $S = 0$. Thus full disclosure is the unique platform-optimal continuation mechanism at every database. Corollary 1 then implies that every equilibrium is efficient.

(ii) $\gamma \geq 1$. When $\gamma \geq 1$, for any feasible pair,

$$\Pi + \gamma S \leq \Pi + \gamma(w(q) - \Pi) = \gamma w(q) + (1-\gamma)\Pi.$$

Because $1-\gamma \leq 0$, the platform's maximal value is attained at the smallest feasible merchant profit, $\Pi = \pi_m(q)$. Thus, for $\gamma > 1$, platform optimality requires the consumer-optimal pair $(\pi_m(q), w(q) - \pi_m(q))$; for $\gamma = 1$, the platform is indifferent over total-surplus-maximizing mechanisms, and the planner-preferred continuation mechanism is again the consumer-optimal one. Hence the planner's reduced continuation payoff in this regime is

$$C(q) \triangleq \pi_m(q) + (1+\gamma)(w(q) - \pi_m(q)) = (1+\gamma)w(q) - \gamma\pi_m(q).$$

The function C is concave because w is linear and π_m, being the maximum of finitely many linear revenue functions, is convex.

Suppose first that the equilibrium involves no disclosure. Under no disclosure the merchant chooses a single price for the whole database. In an equilibrium with $\gamma \geq 1$, this price must be $\underline{\omega}$. To see this, suppose the no-disclosure price were some $a > \underline{\omega}$. Since every type has positive mass in the equilibrium database, consumers with $\omega < a$ would not trade, so the induced total surplus would be strictly below $w(q^*)$. Full disclosure attains total surplus $w(q^*)$. Thus, when $\gamma = 1$, no disclosure at $a > \underline{\omega}$ cannot be platform-optimal. When $\gamma > 1$, the platform can attain the consumer-optimal point with the same minimal merchant profit and strictly higher consumer surplus, so no disclosure at $a > \underline{\omega}$ cannot be platform-optimal either. Hence no disclosure in equilibrium induces price $\underline{\omega}$.

For each price $a \in \Omega$, let $g^a \in \mathbb{R}^\Omega$ denote the revenue vector $g^a(\omega) \triangleq a\mathbb{1}\{\omega \geq a\}$. Then $\pi_m(q) = \max_{a\in\Omega} g^a \cdot q$. Since no disclosure at the lowest price is obedient at q^*, $g^{\underline{\omega}}$ is an active subgradient of the convex function π_m at q^*.

Let $g^* \in \partial\pi_m(q^*)$ satisfy

$$p^* = \gamma\nabla w(q^*) + (1-\gamma)g^*,$$

which is possible because $p^* \in \partial V(q^*)$ and $V(q) = \gamma w(q) + (1-\gamma)\pi_m(q)$ in this regime. Under no disclosure at price $\underline{\omega}$, a type-ω consumer's surplus is $\omega - \underline{\omega} = \nabla w(q^*)(\omega) -$

$g^{\underline{\omega}}(\omega)$. Market clearing gives

$$r_{q^*} = p^* + \nabla w(q^*) - g^{\underline{\omega}}.$$

Combining the last two displays yields

$$r_{q^*} = (1+\gamma)\nabla w(q^*) - \gamma h, \qquad h \triangleq \frac{1}{\gamma} g^{\underline{\omega}} + \frac{\gamma - 1}{\gamma} g^*.$$

Because $\gamma \geq 1$, h is a convex combination of elements of $\partial \pi_m(q^*)$, so $h \in \partial \pi_m(q^*)$. Therefore r_{q^*} is a supergradient of the concave function $C(q) = (1+\gamma)w(q) - \gamma \pi_m(q)$ at q^*. Since $\nabla R(q^*) = -r_{q^*}$, zero belongs to the superdifferential of the concave function $C + R$ at q^*. Hence q^* maximizes $C(q) + R(q)$, and the equilibrium is efficient.

Conversely, suppose the equilibrium does not involve no disclosure. If the equilibrium continuation mechanism is not planner-preferred at q^*, the planner can keep the same database and choose a platform-optimal mechanism with higher welfare, so the equilibrium is inefficient. It remains to consider the case in which the equilibrium implements the consumer-optimal pair at q^*. Then lowest-price no disclosure is not obedient: otherwise, since all prices are at least $\underline{\omega}$, attaining total surplus $w(q^*)$ and merchant profit $\pi_m(q^*) = \underline{\omega} \sum_\omega q^*(\omega)$ would require every consumer to be charged $\underline{\omega}$, so the equilibrium would involve no disclosure, contrary to the maintained assumption. As a result, the uninformed monopoly price is strictly above $\underline{\omega}$. Hence every active price in $\pi_m(q^*)$ is strictly above $\underline{\omega}$, and adding a small mass of the lowest type does not change the revenue of any active uninformed monopoly price. Therefore

$$\pi_m'(q^*; e_{\underline{\omega}}) = 0.$$

At a consumer-optimal mechanism, the lowest type obtains zero consumer surplus. Since $p^* \in \partial V(q^*)$ and the lowest coordinate of every active subgradient of π_m is zero, the equilibrium price for the lowest type is $p^*(\underline{\omega}) = \gamma \underline{\omega}$. Hence $r_{q^*}(\underline{\omega}) = \gamma \underline{\omega}$. The directional derivative of the reduced continuation welfare is

$$C'(q^*; e_{\underline{\omega}}) = (1+\gamma)\underline{\omega} - \gamma \pi_m'(q^*; e_{\underline{\omega}}) = (1+\gamma)\underline{\omega}.$$

Using Lemma A.2,

$$\widehat{W}'(q^*; e_{\underline{\omega}}) = C'(q^*; e_{\underline{\omega}}) - r_{q^*}(\underline{\omega}) = \underline{\omega} > 0.$$

By Proposition A.1, the equilibrium is not locally efficient, and hence not efficient. □

C Proofs for Section 5

C.1 Monopsonist Platform and Data Union

Proof of Remark 1. Let (p^M, q^M, x^M) be a monopsony equilibrium. Since the associated price schedule is finite and each F_ω has strictly positive density on $\mathbb{R}$, q^M is interior. Define

$$A^M \triangleq V(q^M) + \sum_\omega q^M(\omega) U(\omega, x^M).$$

By Lemma A.3, $x^M \in \mathcal{X}(\lambda q^M)$ for every $\lambda > 0$. For λ sufficiently close to one, $\lambda q^M \leq \bar{q}$ because q^M is interior. Along the scaling path λq^M, the monopsonist's profit is

$$\lambda A^M - \sum_\omega \lambda q^M(\omega) r_{\lambda q^M}(\omega).$$

Optimality at $\lambda = 1$ implies that the derivative of this expression is zero. Since

$$\frac{d}{d\lambda} r_{\lambda q^M}(\omega)\Big|_{\lambda=1} = \frac{q^M(\omega)}{\bar{q}(\omega) f_\omega(r_{q^M}(\omega))}$$

we obtain

$$A^M - \sum_\omega q^M(\omega) r_{q^M}(\omega) = \sum_\omega \frac{(q^M(\omega))^2}{\bar{q}(\omega) f_\omega(r_{q^M}(\omega))} > 0.$$

Along the same scaling path, welfare is

$$W(\lambda q^M, x^M) = \lambda A^M + R(\lambda q^M).$$

By Lemma A.2,

$$\frac{d}{d\lambda} W(\lambda q^M, x^M)\Big|_{\lambda=1} = A^M - \sum_\omega q^M(\omega) r_{q^M}(\omega) > 0.$$

Hence, for $\lambda > 1$ sufficiently close to one, $W(\lambda q^M, x^M) > W(q^M, x^M)$. Therefore every monopsony equilibrium is inefficient. □

Proof of Remark 2. Write

$$O(q) \triangleq \sum_\omega \bar{q}(\omega) \int_0^{q(\omega)/\bar{q}(\omega)} F_\omega^{-1}(s)\, ds,$$

so that $O(q) + R(q) = \sum_\omega \bar{q}(\omega) \int_0^1 F_\omega^{-1}(s)\, ds$ is independent of q.

Let (q°, x°) be efficient. We first show that it is induced by some feasible union policy. Since continuation payoffs are finite and F_ω has strictly positive density on $\mathbb{R}$, the efficient database is interior: if $q(\omega) = 0$, adding an arbitrarily small mass of type ω consumers has finite continuation payoff but avoids an arbitrarily low outside-option loss; if $q(\omega) = \bar{q}(\omega)$, removing an arbitrarily small mass avoids an arbitrarily high outside-option cost. Thus $r_{q^\circ}(\omega)$ is finite for every ω.

Define $p^\circ(\omega) = r_{q^\circ}(\omega) - U(\omega, x^\circ)$. Then $p^\circ(\omega) + U(\omega, x^\circ) = r_{q^\circ}(\omega)$, so consumers delegate exactly up to the cutoff that induces q°.

It remains to verify break-even. Let

$$A^\circ \triangleq V(q^\circ) + \sum_\omega q^\circ(\omega) U(\omega, x^\circ).$$

By Lemma A.3, $x^\circ \in \mathcal{X}(\lambda q^\circ)$ for every $\lambda > 0$. Therefore $(\lambda q^\circ, x^\circ)$ is feasible for the planner for all λ near one. Efficiency implies that $\lambda = 1$ locally maximizes $\lambda A^\circ + R(\lambda q^\circ)$. Differentiating at one and using Lemma A.2 gives

$$A^\circ = \sum_\omega q^\circ(\omega) r_{q^\circ}(\omega).$$

Substituting the definition of p° gives $\sum_\omega q^\circ(\omega) p^\circ(\omega) = V(q^\circ)$. Thus (p°, x°) is feasible for the data union's constraints in Section 5.1.

Conversely, every feasible pair (p, x) for the data union induces an allocation $(q_{p,x}, x)$ satisfying $x \in \mathcal{X}(q_{p,x})$, hence feasible for the efficiency benchmark. Under the break-even constraint, and writing $q = q_{p,x}$, the union objective is

$$\begin{aligned}
&\sum_\omega q(\omega)\big(p(\omega) + U(\omega, x)\big) - O(q) \\
&\quad \leq V(q) + \sum_\omega q(\omega) U(\omega, x) - O(q) \\
&\quad = W(q, x) - \sum_\omega \bar{q}(\omega) \int_0^1 F_\omega^{-1}(s)\, ds.
\end{aligned}$$

The first part shows that every efficient allocation is induced by a feasible union policy at which the no-deficit constraint binds, so this upper bound is attained at the efficient welfare level. Hence every data-union optimizer induces an efficient allocation. $\square$

C.2 Contractible actions

If the platform can write an enforceable, ex post observable contract over the merchant's action, obedience is irrelevant. For each ω, let

$$A^C(\omega) \triangleq \arg\max_a (v+\pi)(a,\omega), \qquad \phi^C(\omega) \triangleq \max_a (v+\pi)(a,\omega).$$

The relevant benchmark for the contractible-action economy is

$$\begin{aligned} \max_{q,\, x:\Omega\to\Delta(A)} \quad & \sum_\omega \left(\phi^C(\omega) + \sum_a x(a|\omega)u(a,\omega) \right) q(\omega) + R(q) \\ \text{s.t.} \quad & 0 \le q \le \bar{q}, \\ & x(a|\omega) = 0 \qquad \forall \omega,\ \forall a \notin A^C(\omega). \end{aligned} \tag{C.1}$$

An equilibrium (p^*, q^*, x^*) of this economy is *planner-preferred* if x^* maximizes social welfare among all x satisfying the constraints in (C.1), holding fixed q^*.

Remark C.1. *An allocation (q,x) is induced by a planner-preferred equilibrium of the contractible-action economy if and only if it solves* (C.1).

Proof. For each ω, define

$$m^C(\omega) \triangleq \max_{a\in A^C(\omega)} u(a,\omega).$$

First, let (p^*, q^*, x^*) be a planner-preferred equilibrium of the contractible-action economy. Since action can be assigned typewise, the platform's continuation value is linear: $V^C(q) = \sum_\omega \phi^C(\omega)q(\omega)$. Thus the competitive price of a type-ω record is $\phi^C(\omega)$. Since x^* is planner-preferred, it maximizes consumer surplus among all feasible contracts at the induced database q^*. Because the equilibrium database is interior, this implies

$$\sum_a x^*(a|\omega)u(a,\omega) = m^C(\omega) \qquad \forall \omega.$$

Consumer participation therefore induces the cutoff $r_{q^*}(\omega) = \phi^C(\omega) + m^C(\omega)$. By Lemma A.2, these are exactly the first-order conditions for maximizing the concave function

$$q \mapsto \sum_\omega \left(\phi^C(\omega) + m^C(\omega) \right) q(\omega) + R(q).$$

Therefore q^* solves this maximization problem. Moreover, for every feasible contract x, $\sum_a x(a|\omega)u(a,\omega) \le m^C(\omega)$ for all ω. Hence (q^*, x^*) solves (C.1).

Conversely, let (q°, x°) solve (C.1). Since every feasible contract x satisfies $\sum_a x(a|\omega)u(a,\omega) \leq m^C(\omega)$ for all ω, the database q° solves

$$\max_q \sum_\omega \left(\phi^C(\omega) + m^C(\omega)\right) q(\omega) + R(q) \qquad \text{s.t.} \qquad 0 \leq q \leq \bar{q}.$$

By Lemma A.2, the solution is interior and satisfies $r_{q^\circ}(\omega) = \phi^C(\omega) + m^C(\omega)$. Since (q°, x°) solves (C.1), it must also satisfy $\sum_a x^\circ(a|\omega)u(a,\omega) = m^C(\omega)$ for all ω. Thus x° is planner-preferred at q°. Because $x^\circ(a|\omega) = 0$ whenever $a \notin A^C(\omega)$, the platform is willing to implement x°. At prices $p^\circ(\omega) = \phi^C(\omega)$, the consumer participation cutoffs are $r_{q^\circ}(\omega)$, so the induced database is q°. Hence (q°, x°) is induced by a planner-preferred equilibrium of the contractible-action economy. ◻

C.3 Data Taxes

A type-ω seller now pays a tax $\tau(\omega)$, so her participation condition is $r \leq p(\omega) + U(\omega, x) - \tau(\omega)$.

Remark C.2. *Let (q°, x°) be efficient, let $p^\circ \in \partial V(q^\circ)$, and define*

$$\tau^\circ(\omega) \triangleq p^\circ(\omega) + U(\omega, x^\circ) - r_{q^\circ}(\omega).$$

Then $(p^\circ, q^\circ, x^\circ)$ is an equilibrium of the economy with taxes τ°, and the tax system is budget balanced. If x° admits a differentiable local selection from $\mathcal{X}$, then $\tau^\circ(\omega) = -\varepsilon_{q^\circ}(\omega)$ for every ω.

Proof. Given τ°, a type-ω consumer sells if and only if

$$r \leq p^\circ(\omega) + U(\omega, x^\circ) - \tau^\circ(\omega) = r_{q^\circ}(\omega),$$

so the induced database is q°. Since $p^\circ \in \partial V(q^\circ)$, q° solves the platform's first-period demand problem at price p°. Since $x^\circ \in \mathcal{X}(q^\circ)$, continuation optimality holds. Thus $(p^\circ, q^\circ, x^\circ)$ is an equilibrium of the taxed economy.

Budget balance follows from two identities. First, Lemma A.3 gives $\sum_\omega q^\circ(\omega)p^\circ(\omega) = V(q^\circ)$. Second, the scaling argument in the proof of Remark 2 gives

$$V(q^\circ) + \sum_\omega q^\circ(\omega)U(\omega, x^\circ) = \sum_\omega q^\circ(\omega) r_{q^\circ}(\omega).$$

Combining these identities with the definition of τ° yields

$$\sum_\omega q^\circ(\omega)\tau^\circ(\omega) = V(q^\circ) + \sum_\omega q^\circ(\omega)U(\omega, x^\circ) - \sum_\omega q^\circ(\omega)r_{q^\circ}(\omega) = 0.$$

If x° admits a differentiable local selection $q \mapsto x_q \in \mathcal{X}(q)$ with $x_{q^\circ} = x^\circ$, Lemma A.6 gives $p^\circ = \nabla V(q^\circ)$. Since (q°, x°) is efficient, its derivative along each coordinate direction through this feasible local selection is zero. Therefore

$$0 = \nabla V(q^\circ)(\omega) + U(\omega, x^\circ) - r_{q^\circ}(\omega) + \varepsilon_{q^\circ}(\omega) \qquad \forall\omega.$$

Using $p^\circ = \nabla V(q^\circ)$ and the definition of τ° gives $\tau^\circ(\omega) = -\varepsilon_{q^\circ}(\omega)$. □

C.4 Lindahl Pricing

An allocation is *obedience-constrained efficient* if it solves

$$\begin{aligned}
\max_{q,x:\Omega\to\Delta(A)} \quad & \sum_{a,\omega}(v+\pi+u)(a,\omega)x(a|\omega)q(\omega) + R(q) \\
\text{s.t.} \quad & 0 \le q \le \bar{q}, \\
& \sum_\omega (\pi(a,\omega) - \pi(a',\omega))x(a|\omega)q(\omega) \ge 0 \qquad \forall a, a'.
\end{aligned} \tag{C.2}$$

This benchmark drops platform sequential rationality but keeps merchant obedience. In the Lindahl economy there is a market for each pair (a,ω) with price $p(a,\omega)$. Let $z(a|\omega)$ denote the fraction of type-ω consumers who choose market (a,ω).

Remark C.3. *An allocation is induced by a Lindahl equilibrium if and only if it is obedience-constrained efficient.*

Proof. Write $\chi(a,\omega) = x(a|\omega)q(\omega)$. In terms of χ, the obedience constraints are linear and $q(\omega) = \sum_a \chi(a,\omega)$. The objective in (C.2) is concave in χ, because the continuation term is linear and R is concave in q.

First let $(q^\dagger, x^\dagger)$ be obedience-constrained efficient. Note that $q^\dagger$ must be interior. Define

$$p^\dagger(a,\omega) = r_{q^\dagger}(\omega) - u(a,\omega), \qquad z^\dagger(a|\omega) = \frac{x^\dagger(a|\omega)q^\dagger(\omega)}{\bar{q}(\omega)}.$$

Then $p^\dagger(a,\omega) + u(a,\omega) = r_{q^\dagger}(\omega)$ for every (a,ω). Hence exactly the type-ω consumers with outside option below $r_{q^\dagger}(\omega)$ are willing to participate, and they are indifferent across all markets (a,ω). The choice $z^\dagger$ is therefore individually optimal and clears all Lindahl markets.

It remains to verify platform optimality. Let $\chi^\dagger(a,\omega) = x^\dagger(a|\omega)q^\dagger(\omega)$. Since $\chi^\dagger$ solves the concave problem (C.2), the first-order condition for concave maximization over the feasible set implies that, for every feasible χ satisfying $\sum_a \chi(a,\omega) \le \bar{q}(\omega)$ for every ω,

$$\sum_{a,\omega} \big((v+\pi+u)(a,\omega) - r_{q^\dagger}(\omega)\big)\big(\chi(a,\omega) - \chi^\dagger(a,\omega)\big) \le 0.$$

Using $p^\dagger(a,\omega) = r_{q^\dagger}(\omega) - u(a,\omega)$, this becomes

$$\sum_{a,\omega} \big((v+\pi)(a,\omega) - p^\dagger(a,\omega)\big)\big(\chi(a,\omega) - \chi^\dagger(a,\omega)\big) \le 0.$$

We now show that the same inequality implies platform optimality. Let

$$B(a,\omega) \triangleq (v+\pi)(a,\omega) - p^\dagger(a,\omega).$$

Because $q^\dagger$ is interior, $(1+\varepsilon)\chi^\dagger$ is feasible for (C.2) for all sufficiently small $\varepsilon > 0$. Substituting $\chi = (1+\varepsilon)\chi^\dagger$ in the preceding inequality gives $\sum_{a,\omega} B(a,\omega)\chi^\dagger(a,\omega) \le 0$. Substituting $\chi = 0$ gives the reverse inequality. Hence $\sum_{a,\omega} B(a,\omega)\chi^\dagger(a,\omega) = 0$. Now take any platform-feasible χ, where only obedience and nonnegativity are imposed. For any sufficiently small $\lambda > 0$, $\lambda\chi$ is feasible for (C.2). Substituting $\lambda\chi$ in the preceding inequality and using the displayed equality gives $\lambda \sum_{a,\omega} B(a,\omega)\chi(a,\omega) \le 0$. Thus $\sum_{a,\omega} B(a,\omega)\chi(a,\omega) \le 0$ for every platform-feasible χ, while $\chi^\dagger$ gives value zero. Therefore $\chi^\dagger$ maximizes the platform's Lindahl objective

$$\sum_{a,\omega} \big((v+\pi)(a,\omega) - p^\dagger(a,\omega)\big)\chi(a,\omega)$$

subject to the obedience constraints. Thus $(p^\dagger, z^\dagger, q^\dagger, x^\dagger)$ is a Lindahl equilibrium.

Conversely, let (p^*, z^*, q^*, x^*) be a Lindahl equilibrium and write $\chi^*(a,\omega) = x^*(a|\omega)q^*(\omega)$. Consumer optimality and market clearing imply

$$q^*(\omega) = \bar{q}(\omega)F_\omega\Big(\max_a\{p^*(a,\omega) + u(a,\omega)\}\Big),$$

so $\max_a\{p^*(a,\omega) + u(a,\omega)\} = r_{q^*}(\omega)$. Moreover, if $\chi^*(a,\omega) > 0$, then market clearing requires $z^*(a|\omega) > 0$, and consumer optimality gives equality:

$$p^*(a,\omega) + u(a,\omega) = r_{q^*}(\omega).$$

For unused markets, the left side is weakly below $r_{q^*}(\omega)$.

Take any feasible χ. Platform optimality at prices p^* gives

$$\sum_{a,\omega}\big((v+\pi)(a,\omega)-p^*(a,\omega)\big)\chi(a,\omega) \le \sum_{a,\omega}\big((v+\pi)(a,\omega)-p^*(a,\omega)\big)\chi^*(a,\omega).$$

Since $p^*(a,\omega)+u(a,\omega) \le r_{q^*}(\omega)$ for every (a,ω), the left-hand side is weakly larger than

$$\sum_{a,\omega}\big((v+\pi+u)(a,\omega)-r_{q^*}(\omega)\big)\chi(a,\omega).$$

On the support of χ^*, equality holds. Therefore, for every feasible χ,

$$\sum_{a,\omega}\big((v+\pi+u)(a,\omega)-r_{q^*}(\omega)\big)\big(\chi(a,\omega)-\chi^*(a,\omega)\big) \le 0.$$

This is the first-order condition for maximizing the concave objective in (C.2). Hence (q^*,x^*) is obedience-constrained efficient. □

Online Appendix (For Online Publication Only)

D Omitted Proofs

This section provides proofs omitted from Appendix A. The arguments are variations on standard linear programming and measure-theoretic techniques, adapted to the structure of our platform problem.

We use the joint-allocation variables $\chi(a,\omega) = x(a|\omega)q(\omega)$. In these variables, the platform's continuation problem is

$$\begin{aligned} V(q) = \max_{\chi \geq 0} \quad & \sum_{a,\omega} (v+\pi)(a,\omega)\chi(a,\omega) \\ \text{s.t.} \quad & \sum_{a} \chi(a,\omega) = q(\omega) \qquad \forall \omega, \\ & \sum_{\omega} (\pi(a,\omega) - \pi(a',\omega))\chi(a,\omega) \geq 0 \qquad \forall a, a'. \end{aligned} \tag{D.1}$$

We write $\chi^*(q)$ for the set of optimal solutions of (D.1). If $\chi \in \chi^*(q)$ and $q(\omega) > 0$, the corresponding mechanism is $x(a|\omega) = \chi(a,\omega)/q(\omega)$.

Lemma D.1 (Piecewise linearity of V)**.** *There exist finitely many vectors $a_1, \dots, a_J \in \mathbb{R}^\Omega$ such that*

$$V(q) = \min_{1 \leq j \leq J} a_j \cdot q \qquad \forall q \in \mathbb{R}^\Omega_+.$$

In particular, V is concave, continuous, homogeneous of degree one, and piecewise linear.

Proof. It is enough to prove the claim on the simplex $\Delta(\Omega)$ and then extend it by homogeneity. For each action $a \in A$, let

$$\Delta^*(a) = \left\{ \mu \in \Delta(\Omega) : \sum_{\omega} \mu(\omega)\big(\pi(a,\omega) - \pi(a',\omega)\big) \geq 0 \; \forall a' \in A \right\}.$$

This is a polytope. The graph of the payoff obtained by recommending action a at posteriors for which a is obedient is the polytope

$$E_a = \left\{ (\mu, z) : \mu \in \Delta^*(a),\; z = \sum_{\omega} \mu(\omega)(v+\pi)(a,\omega) \right\}.$$

Let $E = \bigcup_{a \in A} E_a$. By the finite-state persuasion representation, the platform value on the simplex is the upper frontier of the convex hull of E: for every $\mu \in \Delta(\Omega)$,

$$V(\mu) = \max\{z : (\mu, z) \in \mathrm{co}(E)\}.$$

Because E is a finite union of polytopes, $\mathrm{co}(E)$ is a polytope. The upper frontier of a polytope is the pointwise minimum of finitely many affine functions. Hence there are finitely many affine functions $\bar{f}_j(\mu) = \alpha_j \cdot \mu + \beta_j$ such that $V(\mu) = \min_j \bar{f}_j(\mu)$ on $\Delta(\Omega)$.

For $q \in \mathbb{R}^\Omega_+$ with $\sum_\omega q(\omega) > 0$, write $\mu = q / \sum_\omega q(\omega)$. The platform problem is homogeneous: if χ is feasible at q, then $\lambda\chi$ is feasible at λq, and objective values scale by λ. Therefore $V(q) = (\sum_\omega q(\omega))\, V(\mu)$. Define $a_j \in \mathbb{R}^\Omega$ by

$$a_j(\omega) = \alpha_j(\omega) + \beta_j \qquad \forall \omega.$$

Then $(\sum_\omega q(\omega))\, \bar{f}_j(\mu) = a_j \cdot q$, so $V(q) = \min_j a_j \cdot q$ on $\mathbb{R}^\Omega_+ \setminus \{0\}$. The equality also holds at $q = 0$ by homogeneity. Concavity and continuity follow because V is a finite minimum of linear functions, and piecewise linearity follows from the same representation. □

After deleting duplicate linear pieces, we regard the vectors $a_1, \dots, a_J$ as distinct. For each j, let

$$R_j \triangleq \{q \in \mathbb{R}^\Omega_+ : V(q) = a_j \cdot q\}.$$

Lemma D.2 (Differentiability regions)**.** *Let $q \in \mathbb{R}^\Omega_{++}$. If V is differentiable at q, then there is an index j and an open neighborhood N of q in $\mathbb{R}^\Omega$ such that $V(q') = a_j \cdot q'$ for every $q' \in N \cap \mathbb{R}^\Omega_+$. Conversely, if two distinct linear pieces are active at q, then V is not differentiable at q. Consequently, V is differentiable on $\mathbb{R}^\Omega_{++}$ outside a finite union of affine hyperplanes.*

Proof. Since $V = \min_j a_j \cdot q$, the superdifferential of V at an interior point q is the convex hull of the active gradients:

$$\partial V(q) = \mathrm{co}\{a_j : V(q) = a_j \cdot q\}.$$

If two distinct pieces a_i and a_j are active at q, then $\partial V(q)$ contains both a_i and a_j, so it is not a singleton. Hence V is not differentiable at q.

Suppose now that V is differentiable at q. Since duplicate pieces have been deleted, the preceding paragraph implies that exactly one index, say j, is active at q. For every $i \neq j$,

$$a_j \cdot q < a_i \cdot q.$$

There are finitely many such inequalities, so they continue to hold in some open neighborhood N of q. Thus $V(q') = a_j \cdot q'$ on $N \cap \mathbb{R}^{\Omega}_{+}$, and V is locally linear there. Finally, nondifferentiability can occur only at points where two distinct pieces are active, and each equality $(a_i - a_j) \cdot q = 0$ is an affine hyperplane. There are finitely many pairs (i, j), so the set of nondifferentiability points is contained in a finite union of affine hyperplanes. □

For the next proofs, we recall the standard finite-basis argument for linear programs. Consider a linear program in standard form,

$$\max\{\bar{c} \cdot z : Az = b,\ z \geq 0\},$$

where A is a fixed matrix. We first delete redundant rows of A, so A has full row rank r. A *basis* is a set $\mathfrak{b}$ of r columns of A such that the square matrix $A_{\mathfrak{b}}$ is invertible. The associated basic solution is obtained by setting all nonbasic variables equal to zero and solving $z_{\mathfrak{b}} = A_{\mathfrak{b}}^{-1} b$. When b depends linearly on a parameter, each basic solution is affine in that parameter. Since there are only finitely many bases, a parametric linear program has only finitely many basic affine candidate solutions.

Proof of Lemma A.4. Let

$$P(q) \triangleq \max_{x \in \mathcal{X}(q)} \sum_{a,\omega} (v + \pi + u)(a, \omega) x(a|\omega) q(\omega)$$

be the planner-preferred continuation value, so that $\widehat{W}(q) = P(q) + R(q)$. We prove that P is continuous and piecewise affine on $\mathbb{R}^{\Omega}_{++}$.

Fix j. On R_j, the platform value is $a_j \cdot q$. Hence, for $q \in R_j$, the value $P(q)$ equals the value of the auxiliary program

$$\begin{aligned} P_j(q) = \max_{\chi \geq 0} \quad & \sum_{a,\omega} (v + \pi + u)(a, \omega) \chi(a, \omega) \\ \text{s.t.} \quad & \sum_{a} \chi(a, \omega) = q(\omega) \qquad \forall \omega, \\ & \sum_{\omega} (\pi(a, \omega) - \pi(a', \omega)) \chi(a, \omega) \geq 0 \qquad \forall a, a', \\ & \sum_{a,\omega} (v + \pi)(a, \omega) \chi(a, \omega) = a_j \cdot q. \end{aligned} \tag{D.2}$$

The last equality is exactly the condition that χ be platform-optimal at q, because $V(q) = a_j \cdot q$ on R_j.

Introduce one nonnegative slack variable $s_{aa'}$ for each obedience inequality, rewriting that inequality as an equality plus a slack. Let $z = (\chi, s)$ denote the vector consisting of the joint-allocation variables and all slack variables. Then (D.2) can be written in standard form as

$$\max\{\bar{d} \cdot z : Az = H_j q,\ z \geq 0\}, \tag{D.3}$$

where A is a fixed matrix, H_j is the matrix collecting the right-hand sides q and $a_j \cdot q$, and $\bar{d}$ has coefficients $(v + \pi + u)(a, \omega)$ on the χ-coordinates and zero on the slack coordinates.

Let $\mathcal{B}_j$ be the finite set of bases of (D.3). For a basis $\mathfrak{b} \in \mathcal{B}_j$, let $z^{\mathfrak{b}}(q)$ be the corresponding basic solution. Because the right-hand side is $H_j q$, the map $q \mapsto z^{\mathfrak{b}}(q)$ is affine. Let

$$g_{\mathfrak{b}}(q) \triangleq \bar{d} \cdot z^{\mathfrak{b}}(q).$$

The basis $\mathfrak{b}$ determines a dual vector

$$\lambda_{\mathfrak{b}}^{\top} \triangleq \bar{d}_{\mathfrak{b}}^{\top} A_{\mathfrak{b}}^{-1}.$$

We call $\mathfrak{b}$ dual feasible if $\lambda_{\mathfrak{b}}^{\top} A \geq \bar{d}^{\top}$. If $\mathfrak{b}$ is dual feasible and $z^{\mathfrak{b}}(q) \geq 0$, then $z^{\mathfrak{b}}(q)$ is optimal: for every feasible z,

$$\bar{d} \cdot z \leq \lambda_{\mathfrak{b}}^{\top} A z = \lambda_{\mathfrak{b}}^{\top} H_j q = \bar{d}_{\mathfrak{b}}^{\top} A_{\mathfrak{b}}^{-1} H_j q = \bar{d} \cdot z^{\mathfrak{b}}(q) = g_{\mathfrak{b}}(q).$$

Conversely, if the auxiliary program has an optimum at q, then some optimal solution is a basic feasible solution; the associated basis is feasible at q and satisfies the dual optimality inequalities. Thus, for every $q \in R_j$, the value $P_j(q)$ is equal to $g_{\mathfrak{b}}(q)$ for at least one dual-feasible basis $\mathfrak{b}$ whose basic solution is feasible at q.

For every dual-feasible basis $\mathfrak{b}$, define

$$S_{j,\mathfrak{b}} \triangleq \{q \in R_j : z^{\mathfrak{b}}(q) \geq 0\}.$$

Since $z^{\mathfrak{b}}(q)$ is affine in q, $S_{j,\mathfrak{b}}$ is a polyhedron. The preceding paragraph shows that the finitely many sets $S_{j,\mathfrak{b}}$ cover R_j and that $P_j(q) = g_{\mathfrak{b}}(q)$ on $S_{j,\mathfrak{b}}$. If $q \in S_{j,\mathfrak{b}} \cap S_{j,\mathfrak{b}'}$, then both basic solutions are optimal for the same program at q, so

$$g_{\mathfrak{b}}(q) = P_j(q) = g_{\mathfrak{b}'}(q).$$

Therefore the affine formulas agree on overlaps.

We now prove continuity. The sets $S_{j,\mathfrak{b}}$, over all j and all dual-feasible bases $\mathfrak{b}$, form a finite cover of $\mathbb{R}^{\Omega}_{++}$. On each such set, P agrees with the affine function $g_{\mathfrak{b}}$, and these affine functions agree at every point where their domains overlap. Let $q_n \to q$ with $q_n, q \in \mathbb{R}^{\Omega}_{++}$. For each n, choose one pair $(j_n, \mathfrak{b}_n)$ such that $q_n \in S_{j_n,\mathfrak{b}_n}$. Since there are finitely many pairs, every subsequence has a further subsequence, still denoted q_n, for which $(j_n, \mathfrak{b}_n) = (j, \mathfrak{b})$ is constant. The set $S_{j,\mathfrak{b}}$ is closed relative to $\mathbb{R}^{\Omega}_{++}$, because it is defined by weak linear inequalities together with the equalities defining R_j. Hence $q \in S_{j,\mathfrak{b}}$. Therefore

$$P(q_n) = g_{\mathfrak{b}}(q_n) \to g_{\mathfrak{b}}(q) = P(q).$$

Thus every subsequence of $P(q_n)$ has a further subsequence converging to $P(q)$, which implies $P(q_n) \to P(q)$. Hence P is continuous.

Taking the common refinement of the finitely many polyhedra $S_{j,\mathfrak{b}}$ gives a finite polyhedral subdivision of $\mathbb{R}^{\Omega}_{++}$. On each relatively open cell of this subdivision, one affine formula $g_{\mathfrak{b}}$ applies and all other applicable formulas agree with it. Thus P is piecewise affine.

Next, fix an interior database q and a direction d. Because the subdivision is finite, the line segment $q + td$ crosses only finitely many cell boundaries near $t = 0$. Hence, for some $\varepsilon > 0$, the restriction of P to $q + td$ is affine on $(0, \varepsilon)$. Therefore the right directional derivative $P'(q; d)$ exists. Since R is differentiable on the interior domain, $\widehat{W} = P + R$ is directionally differentiable there.

Finally, suppose that $\widehat{W}'(q; d) \leq 0$ for every direction d. Since the subdivision has finitely many boundary hyperplanes, there is a neighborhood N of q such that for every $q + d \in N$, the segment $\{q + td : 0 < t \leq 1\}$ lies in a single cell, or in a common face on which the adjacent affine formulas agree. Hence

$$P(q + d) - P(q) = P'(q; d)$$

for every sufficiently small d. By concavity and differentiability of R,

$$R(q + d) - R(q) \leq \nabla R(q) \cdot d.$$

Thus

$$\widehat{W}(q + d) - \widehat{W}(q) \leq P'(q; d) + \nabla R(q) \cdot d = \widehat{W}'(q; d) \leq 0.$$

So q is a local maximizer of $\widehat{W}$. □

Proof of Lemma A.7. Let q^* be regular. By Lemma D.2, there is an index j and an open neighborhood N of q^* such that $V(q) = a_j \cdot q$ for every $q \in N$. Shrinking N if necessary, assume $N \subset \mathbb{R}^{\Omega}_{++}$ and N is convex.

We first show that the optimal joint allocation is unique on a possibly smaller neighborhood of q^*. Suppose not. Then there exist $q_n \to q^*$ and two distinct optimal joint allocations $\chi_n^1, \chi_n^2 \in \chi^*(q_n)$. For large n, the reflected database $\tilde{q}_n = 2q^* - q_n$ also lies in N. Choose any $\tilde{\chi}_n \in \chi^*(\tilde{q}_n)$. For $i = 1, 2$, the average $(\chi_n^i + \tilde{\chi}_n)/2$ is feasible at q^*, because all feasibility constraints in (D.1) are linear in (χ, q). Its objective value is

$$\frac{1}{2}V(q_n) + \frac{1}{2}V(\tilde{q}_n) = \frac{1}{2}a_j \cdot q_n + \frac{1}{2}a_j \cdot \tilde{q}_n = a_j \cdot q^* = V(q^*).$$

Thus the two averages are optimal at q^*. Since $\chi_n^1 \neq \chi_n^2$, the two averages are distinct, contradicting uniqueness at the regular database q^*. Hence there is a neighborhood $N' \subset N$ of q^* on which the optimal joint allocation is unique.

Now take any $q_1, q_2 \in N'$ and $\lambda \in [0, 1]$ such that $q_\lambda = \lambda q_1 + (1 - \lambda)q_2$ also lies in N'. Let χ_{q_i} be the unique optimal allocation at q_i. The convex combination $\lambda\chi_{q_1} + (1 - \lambda)\chi_{q_2}$ is feasible at q_λ and has objective value

$$\lambda V(q_1) + (1 - \lambda)V(q_2) = \lambda a_j \cdot q_1 + (1 - \lambda)a_j \cdot q_2 = a_j \cdot q_\lambda = V(q_\lambda).$$

By uniqueness at q_λ, this convex combination equals the unique optimal allocation χ_{q_λ}. Therefore $q \mapsto \chi_q$ is affine on N'. Since $q^*(\omega) > 0$ for every ω, after shrinking N' all coordinates $q(\omega)$ are bounded away from zero on N'. Hence

$$x_q(a|\omega) = \frac{\chi_q(a, \omega)}{q(\omega)}$$

is well defined and differentiable on N'. This proves the lemma. □

Lemma D.3 (Generic platform regularity)**.** *For every fixed π and Lebesgue-a.e. v, ($\mathcal{P}_q$) has a unique optimal joint allocation at every interior database.*

Proof. Let $N = |A||\Omega|$ be the dimension of the joint-allocation space. For each database q, the feasible set of (D.1) is a polytope in $\mathbb{R}^N$. Its defining constraints are the nonnegativity constraints, the obedience inequalities, and the database equalities. The normal vectors of all these constraints are fixed by π and do not depend on q.

Suppose that, for some objective vector $c \in \mathbb{R}^{A \times \Omega}$ and some interior database q, the maximizer of $c \cdot \chi$ over the feasible polytope is not unique. Then the optimal face has positive dimension. Choose a point in the relative interior of that face. Let I be the set of constraints that bind at that point. The tangent space of the face is

$$T_I = \{h : n_\ell \cdot h = 0 \text{ for every binding normal } n_\ell, \sum_a h(a, \omega) = 0 \ \forall \omega\}.$$

Since the face has positive dimension, T_I is nontrivial. Because the entire face is optimal, the objective is constant along every tangent direction in the face. Hence $c \cdot h = 0, \forall h \in T_I$. Therefore $c \in T_I^{\perp}$. Since T_I is nontrivial, $T_I^{\perp}$ is a proper linear subspace of $\mathbb{R}^{A \times \Omega}$.

There are only finitely many possible binding sets I. For each binding set with positive-dimensional tangent space, the condition that c vanish on some nonzero direction in that tangent space means that c belongs to the proper subspace $T_I^{\perp}$. Thus the set of objective vectors that yield nonuniqueness at some interior database is contained in a finite union of proper linear subspaces of $\mathbb{R}^{A \times \Omega}$, and therefore has Lebesgue measure zero. In our problem $c = v + \pi$. Since π is fixed, the same null-set conclusion holds for Lebesgue-a.e. v. □

Lemma D.4 (Finite stratification)**.** *Suppose the platform payoffs are regular. Then there is a finite collection of relatively open polyhedral sets $\{C_j\}_{j=1}^{J}$ covering $\mathbb{R}^{\Omega}_{++}$ such that, on each C_j, the unique optimal joint allocation is affine in q. Moreover, for*

$$\Gamma_j = \{(q, p) : q \in C_j, \ p \in \partial V(q)\},$$

the set Γ_j is contained in a finite union of polyhedra of dimension at most $|\Omega|$.

Proof. Introduce slack variables for the obedience inequalities in (D.1). Let $z = (\chi, s)$ collect the joint-allocation variables and these slack variables. The platform problem can be written in standard form as

$$\max\{\bar{c} \cdot z : Az = Hq, \ z \geq 0\},$$

where A and H are fixed matrices and $\bar{c}$ has coefficients $(v + \pi)(a, \omega)$ on the χ-coordinates and zero on the slack coordinates.

Let $\mathcal{B}$ be the finite set of bases of this standard-form program. For a basis $\mathfrak{b} \in \mathcal{B}$, let $z^{\mathfrak{b}}(q)$ be its basic solution. The map $q \mapsto z^{\mathfrak{b}}(q)$ is affine. Let $\mu_{\mathfrak{b}}^{\top} = \bar{c}_{\mathfrak{b}}^{\top} A_{\mathfrak{b}}^{-1}$. If $\mu_{\mathfrak{b}}^{\top} A \geq \bar{c}^{\top}$, then $\mathfrak{b}$ is dual feasible; whenever $z^{\mathfrak{b}}(q) \geq 0$, the corresponding basic solution is optimal. Conversely, every optimum has an optimal basic feasible solution. Hence the finitely many polyhedra

$$T_{\mathfrak{b}} = \{q \in \mathbb{R}^{\Omega}_{++} : z^{\mathfrak{b}}(q) \geq 0\},$$

as $\mathfrak{b}$ ranges over dual-feasible bases, cover $\mathbb{R}^{\Omega}_{++}$.

Take the finite family of hyperplanes that form the boundaries of the polyhedra $T_{\mathfrak{b}}$ and of the value regions R_i from Lemma D.1. The relative interiors of all nonempty faces generated by this finite hyperplane arrangement form a finite collection of relatively open polyhedral cells covering $\mathbb{R}^{\Omega}_{++}$. Denote these cells by $C_1, \ldots, C_J$.

Fix one cell C_j. For each dual-feasible basis $\mathfrak{b}$, either $C_j \subseteq T_{\mathfrak{b}}$ or $C_j \cap T_{\mathfrak{b}} = \varnothing$, because the cell is a relative interior of the common refinement of the boundary hyperplanes. Thus the set of feasible dual-feasible bases is constant on C_j. Each such basis gives an affine formula for an optimal joint allocation. Since platform payoffs are regular, the optimal joint allocation is unique at every interior database, so all feasible optimal basis formulas coincide on C_j. Therefore there is a matrix M_j such that the unique optimal joint allocation satisfies $\chi(q) = M_j q$ for every $q \in C_j$.

It remains to prove the dimension bound for Γ_j. Let $k = \dim C_j$, and let L_j be the linear subspace parallel to the affine hull of C_j. Recall from Lemma D.1 that $V(q) = \min_i a_i \cdot q$. For a point q, call an index i active if $V(q) = a_i \cdot q$. Because the cells C_j refine the boundaries of the regions $R_i = \{q : V(q) = a_i \cdot q\}$, for each i either $C_j \subseteq R_i$ or $C_j \cap R_i = \varnothing$. Hence the set of active indices is constant on C_j; denote it by I_j.

Fix any $i_0 \in I_j$. If $i \in I_j$, then $a_i \cdot q = a_{i_0} \cdot q$ for every $q \in C_j$. Therefore, for every direction $d \in L_j$, we have $(a_i - a_{i_0}) \cdot d = 0$. Thus $a_i - a_{i_0} \in L_j^{\perp}$, and all vectors $\{a_i : i \in I_j\}$ lie in the affine set $a_{i_0} + L_j^{\perp}$, whose dimension is $|\Omega| - k$.

For a polyhedral concave function written as a minimum of affine functions, the superdifferential at q is the convex hull of the active coefficient vectors:

$$\partial V(q) = \mathrm{co}\{a_i : V(q) = a_i \cdot q\}.$$

Hence, for every $q \in C_j$, every $p \in \partial V(q)$ belongs to $a_{i_0} + L_j^{\perp}$. Therefore Γ_j is contained in $C_j \times (a_{i_0} + L_j^{\perp})$. The first factor has dimension k, and the second has dimension at most $|\Omega| - k$. Thus Γ_j is contained in a polyhedral set of dimension at most $|\Omega|$. □

Before proceeding to the proof of Lemma A.9, we introduce a corollary of Lemma A.8 which provides an easy condition to check for inefficiency.

Corollary D.1 (A direct inefficiency test)**.** *If expression* (A.1) *is nonzero for some used action a, the equilibrium is inefficient.*

Proof. If the derivative in Lemma A.8 is positive, then moving in the positive t direction raises welfare for small $t > 0$. If it is negative, moving in the negative t direction raises welfare for small $|t|$. In either case the equilibrium is not locally efficient, hence not efficient. $\square$

Proof of Lemma A.9. Fix regular platform payoffs and write $n = |\Omega|$ and $m = |A|$. By Lemma D.4, the interior database space is covered by finitely many cells C_j. On each C_j, the unique optimal joint allocation is $\chi_j(q) = M_j q$, and the induced mechanism is $x_j(a|\omega;q) = \chi_j(a,\omega;q)/q(\omega)$.

We first identify the candidate triples (u,q,p) that can arise from locally efficient nondeterministic equilibria. Take any such equilibrium. Its database q belongs to some cell C_j. Since the mechanism is nondeterministic, there exist an action a and a type ω_0 such that $0 < x_j(a|\omega_0;q) < 1$. Corollary D.1 then implies that the segment-scaling derivative for action a must vanish. Thus the equilibrium generates a triple (u,q,p) in one of the sets defined below. Fix a cell C_j, an action a, and a type ω_0. Let

$$L_{j,a}(u,q) = \sum_{\omega} x_j(a|\omega;q)q(\omega)\big(u(a,\omega) - U(\omega, x_j(q))\big).$$

Let S_{j,a,ω_0} be the set of triples (u,q,p) such that $q \in C_j$, $p \in \partial V(q)$, $0 < x_j(a|\omega_0;q) < 1$, and $L_{j,a}(u,q) = 0$. It is enough to show that the implied set of market-clearing $(u,\bar{q})$ of each S_{j,a,ω_0} has Lebesgue measure zero.

For fixed q, the map $u \mapsto L_{j,a}(u,q)$ is linear. The coefficient on $u(a,\omega_0)$ is $\beta_{j,a,\omega_0}(q) = x_j(a|\omega_0;q)q(\omega_0)(1 - x_j(a|\omega_0;q))$, which is strictly positive on S_{j,a,ω_0}.

By Lemma D.4, the graph $\Gamma_j = \{(q,p) : q \in C_j,\ p \in \partial V(q)\}$ is contained in a finite union of polyhedra $\Gamma_{j\ell}$, each of dimension at most n. Fix one such polyhedron and write $d_{j\ell} \leq n$ for its dimension. Since $\Gamma_{j\ell}$ is a polyhedron of dimension $d_{j\ell}$, after choosing coordinates on its affine hull, points in $\Gamma_{j\ell}$ can be written as $(q(y), p(y))$, where y ranges over a subset of $\mathbb{R}^{d_{j\ell}}$, and the map $y \mapsto (q(y), p(y))$ is affine.

For each integer $N \geq 1$, restrict attention to the part of S_{j,a,ω_0} satisfying $(q,p) \in \Gamma_{j\ell}$, $\|u\|_\infty \leq N$, $\|(q,p)\|_\infty \leq N$, $q(\omega) \geq 1/N$ for every ω, and $\beta_{j,a,\omega_0}(q) \geq 1/N$. The union of these restricted sets over all N and over the finitely many ℓ covers S_{j,a,ω_0}, because every point in S_{j,a,ω_0} has $q(\omega) > 0$ for all ω and $\beta_{j,a,\omega_0}(q) > 0$.

On one such restricted set, write $(q,p) = (q(y), p(y))$. The functions $y \mapsto x_j(b|\omega; q(y))$ are Lipschitz because their numerators are linear in $q(y)$ and their denominators $q(\omega; y)$ are bounded below by $1/N$. Hence the coefficients of the linear equation $L_{j,a}(u, q(y)) = 0$ are Lipschitz functions of y. Since the coefficient on $u(a, \omega_0)$ is bounded below by $1/N$, this equation solves $u(a, \omega_0)$ as a Lipschitz function of the remaining $mn-1$ coordinates of u and of the parameter y. Therefore each restricted set is contained in the image of a Lipschitz map whose domain has dimension at most $(mn-1) + d_{j\ell} \leq mn + n - 1$.

Define the market-clearing map by

$$T_j(u,q,p) = \left(u, \left(\frac{q(\omega)}{F_\omega(p(\omega) + U(\omega, x_j(q)))}\right)_{\omega\in\Omega}\right).$$

If (u,q,p) comes from an equilibrium, then market clearing implies that the associated primitive pair $(u, \bar{q})$ is exactly $T_j(u,q,p)$.

On each restricted set above, T_j is Lipschitz as a function of the variables (u,q,p). Indeed, $q \mapsto x_j(q)$ is Lipschitz because all coordinates of q are bounded away from zero. Since u is bounded, $(u,q) \mapsto U(\omega, x_j(q))$ is Lipschitz. Thus $(u,q,p) \mapsto p(\omega) + U(\omega, x_j(q))$ is Lipschitz and has range contained in a compact interval. On this interval, F_ω is Lipschitz because it has bounded density, and F_ω is bounded away from zero because it is strictly positive at every finite argument. Hence the reciprocal of F_ω is Lipschitz on the same interval, and every coordinate of T_j is Lipschitz.

Let $D = mn + n$, the dimension of the primitive space $(u, \bar{q})$. A bounded subset of $\mathbb{R}^{D-1}$ has Lebesgue-null image under any Lipschitz map into $\mathbb{R}^D$. Since each restricted set is contained in a Lipschitz image of dimension at most $D-1$ and T_j is Lipschitz on that set, the image of each restricted set under T_j has D-dimensional Lebesgue measure zero. Taking the countable union over N and the finite union over ℓ shows that $T_j(S_{j,a,\omega_0})$ is null. Finally, taking the finite union over j, a, and ω_0 proves that the set of primitives admitting a locally efficient equilibrium with a nondeterministic mechanism has Lebesgue measure zero. $\square$

E Supplementary Examples

E.1 Necessity of Differentiability for Proposition A.3

Proposition A.3 shows that uniqueness of x^* is not necessary for the deterministic-efficiency conclusion. What is necessary is differentiability, or more economically, local stability of the selected mechanism. The next example shows why that condition cannot simply be dropped.

Example E.1 (Why differentiability is needed)**.** A deterministic equilibrium need not be locally efficient absent a differentiable local selection. Let $\Omega = \{L, H\}$, $A = \{a_L, a_M, a_H\}$, with merchant payoffs

$$\pi(a_L, L) = 2, \quad \pi(a_M, L) = \pi(a_M, H) = 1, \quad \pi(a_H, H) = 2,$$

and the other two type-action payoffs zero, so a_M is obedient only at the posterior with equal mass on L and H. Let the effective payoff $v + \pi$ equal one at (a_M, L), (a_M, H), (a_H, H) and zero otherwise.

At $q^* = (1, 1)$ the unique optimal joint allocation pools both types at a_M, a deterministic mechanism. The platform value is

$$V(q) = \min\{2q(H), q(L) + q(H)\}.$$

Indeed, the platform can obtain $q(L) + q(H)$ by pooling equal masses of L and H at a_M and sending any excess H-mass to a_H, while it cannot obtain more than $2q(H)$ because every payoff-relevant recommendation requires either an H record or a balanced a_M pool. Hence V has a kink at q^*.

Take $p^* = (1, 1) \in \partial V(q^*)$ and choose $\bar{q}, F$ so that market clearing holds at q^*. Let consumer payoffs from a_M be zero and let $u(a_H, H) > 0$. If the mass of H-records is increased slightly, the platform keeps the original a_M pool intact and assigns the marginal H-mass to a_H. The platform value increases at rate one, the aggregate outside-option term decreases at rate one, and consumer surplus increases at rate $u(a_H, H)$. Thus welfare increases. The deterministic equilibrium at q^* is locally inefficient, precisely because no differentiable optimal selection exists at the kink of V.

E.2 Imperfect Data Records

This subsection shows how imperfect records introduce a standard adverse-selection distortion. Suppose that consumers know their payoff-relevant type ω, but the traded record is only a verifiable signal s. The platform can buy and condition on s, but not on ω. To keep the notation minimal, consider one signal s and two latent payoff types, H and L. Let $\bar{q}_H, \bar{q}_L > 0$ denote the masses of the two latent types with signal s, and let F_H, F_L denote their outside-option distributions. Assume that there is only one action, and the platform's value from a participating type-H consumer is one, its value from a participating type-L consumer is zero, and consumer trading surplus is zero.

At a signal price p, the masses of sellers are $m_H(p) = \bar{q}_H F_H(p)$ and $m_L(p) = \bar{q}_L F_L(p)$. In a competitive market for anonymous s-records, a price-taking platform buys a representative record from the pool of consumers already willing to sell at price p. Therefore its marginal value is the average value of that pool. A competitive equilibrium must satisfy

$$p = \frac{m_H(p)}{m_H(p) + m_L(p)}.$$

This condition already accounts for selection: the platform understands that the seller pool at price p has composition $(m_H(p), m_L(p))$.

A planner facing the same informational restriction cannot condition transfers on the latent type either. The planner can choose the same anonymous signal price p, and consumers self-select into participation exactly as in the market. The difference is the margin considered by the planner. When the planner raises p, the additional records come from the marginal entrants, whose composition is proportional to $(m'_H(p), m'_L(p))$. Hence the planner's first-order condition is

$$p = \frac{m'_H(p)}{m'_H(p) + m'_L(p)}.$$

Thus efficiency requires the average composition of the current seller pool to coincide with the composition of the marginal entrants. This is a knife-edge condition:

$$\frac{m_H(p)}{m_H(p) + m_L(p)} = \frac{m'_H(p)}{m'_H(p) + m'_L(p)}.$$

This equality can only be satisfied for special primitives. When it fails, the competitive price differs from the efficient anonymous signal price.

For a concrete example, take $\bar{q}_H = \bar{q}_L = 1$, $F_H(p) = p$, and $F_L(p) = p^2$ on $[0, 1]$. Then $m_H(p) = p$ and $m_L(p) = p^2$. The competitive condition is $p = 1/(1 + p)$, with solultion solution $p^{CE} = (\sqrt{5} - 1)/2$. The planner condition is $p = 1/(1 + 2p)$, with solution $p^{SP} = 1/2$. Hence the competitive allocation is inefficient.

This example shows that imperfect records introduce the standard adverse-selection wedge: market prices reflect the average value of the current selected pool, whereas efficiency depends on the marginal consumers induced to participate. The main text assumes this force away by taking the traded record to reveal ω, thereby isolating the data externality generated by endogenous data use.

F Multiple Competing Platforms

This appendix extends the competitive data economy of Section 2 and the results of Section 3 in two directions. First, it drops the representative-platform assumption and allows for finitely many heterogeneous platforms, each with a possibly overlapping set of finitely many merchants. This shows that the representative-platform formulation is immaterial for the main results. Second, it relaxes the assumption that platforms have full bargaining power vis-à-vis merchants. Platforms may no longer extract merchants' entire profits through access fees. By modifying the efficiency benchmark accordingly, the appendix shows that these simplifying assumptions are not substantive and serve only to make the analysis in the main text cleaner.

Model and Equilibrium. Let $\mathcal{K} = \{1, \ldots, K\}$ be a finite set of platforms. Platform k has a finite set $I_k = \{1, \ldots, n_k\}$ of merchants. A consumer can sell the exclusive right to use her data record to at most one platform. Under this assumption, it is immaterial whether the same merchant operates on multiple platforms. In particular, merchants may multi-home across platforms, provided their payoffs are additive across platforms and there are no capacity constraints.

For each platform k and merchant $i \in I_k$, let $A_{k,i}$ denote the finite action set of merchant i, and let $A_k \triangleq \prod_{i \in I_k} A_{k,i}$ denote the set of downstream action profiles on platform k. If a type-ω consumer participates on platform k and action profile $a_k \in A_k$ is implemented, the consumer obtains trading surplus $u_k(a_k, \omega)$, platform k obtains direct payoff $v_k(a_k, \omega)$, and merchant i obtains gross payoff $\pi_{k,i}(a_k, \omega)$.

Given a database $q_k \in \mathbb{R}^{\Omega}_{+}$, platform k chooses a public recommendation mechanism $x_k : \Omega \to \Delta(A_k)$.[18] The value of database q_k for platform k is

$$
\begin{aligned}
V_k(q_k) \triangleq \max_{x_k:\Omega\to\Delta(A_k)} \quad & \sum_{\omega,a_k} v_k(a_k,\omega)\, x_k(a_k|\omega) q_k(\omega) \\
\text{such that} \quad & \sum_{\omega} q_k(\omega) x_k(a_k|\omega) \Big[\pi_{k,i}(a_k,\omega) - \pi_{k,i}(a'_{k,i}, a_{k,-i}, \omega)\Big] \geq 0, \forall\, i, a_k, a'_{k,i}.
\end{aligned}
$$

Let $\mathcal{X}_k(q_k)$ denote the set of maximizers for this problem. Since the downstream environment is finite, this is a finite linear program. Thus V_k is homogeneous of degree one, concave, continuous, and piecewise linear, as in the main text. For $x_k \in \mathcal{X}_k(q_k)$, define type-$\omega$ consumer surplus on platform k by $U_k(\omega, x_k) \triangleq \sum_{a_k} x_k(a_k|\omega) u_k(a_k,\omega)$.

Each consumer can sell her record to at most one platform (*exclusivity*). Let $\alpha_0(\omega, r)$ and $\alpha_k(\omega, r)$ denote the probability that a type-ω consumer with outside option r keeps the record or sells it to platform k, respectively. Thus, exclusivity requires that $\alpha_0(\omega,r) + \sum_{k\in\mathcal{K}} \alpha_k(\omega, r) = 1$ for all ω and r. We maintain this assumption in the next definition.

Definition F.1. *A profile $\big((p^*_k, q^*_k, x^*_k)_{k\in\mathcal{K}}, \alpha^*\big)$ is an equilibrium of the K-platform competitive economy if the following conditions hold.*

1. *Given p^*_k, each platform k solves its first-period demand problem, that is,*
$$q^*_k \in \arg\max_{q_k\in\mathbb{R}^{\Omega}_{+}} \left\{ V_k(q_k) - \sum_{\omega} p^*_k(\omega) q_k(\omega) \right\} \qquad \text{for all } k \in \mathcal{K}.$$
2. *Given q^*_k, each platform k uses its database optimally, that is, $x^*_k \in \mathcal{X}_k(q^*_k)$ for all k.*
3. *Given $(p^*_k, x^*_k)_{k\in\mathcal{K}}$, consumers are assigned optimally. Let $B^*_k(\omega) \triangleq p^*_k(\omega) + U_k(\omega, x^*_k)$ denote the value obtained by a type-ω consumer from selling her record to platform k. For every $k \in \mathcal{K}$, $\alpha^*_k(\omega, r) > 0$ implies that $B^*_k(\omega) \geq r$ and $B^*_k(\omega) \geq B^*_\ell(\omega)$ for all $\ell \in \mathcal{K}$. Moreover, $\alpha^*_0(\omega, r) > 0$ implies $r \geq \max_{\ell\in\mathcal{K}} B^*_\ell(\omega)$.*
4. *Data markets clear, that is, $q^*_k(\omega) = \bar{q}(\omega) \int \alpha^*_k(\omega, r) dF_\omega(r)$ for all k and ω.*

Letting $M^*(\omega) \triangleq \max_k B^*_k(\omega)$ and $Q^*(\omega) \triangleq \sum_k q^*_k(\omega)$, market clearing can be equivalently written as $Q^*(\omega) = \bar{q}(\omega) F_\omega(M^*(\omega))$ for all ω. The proof of equilibrium existence is

[18] We restrict attention to public recommendation mechanisms as that is important for establishing the analogue of Proposition 2. This result would also extend to certain cases of private information provision.

the same as in the one-platform economy and is omitted. Since each V_k is homogeneous, each platform earns zero profit in equilibrium. Moreover, $p_k^* \in \partial V_k(q_k^*)$ for all $k \in \mathcal{K}$. If V_k is differentiable at q_k^*, then $p_k^*(\omega) = \partial V_k(q_k^*)/\partial q_k(\omega)$ for all $k \in \mathcal{K}$ and $\omega \in \Omega$.

We say that an equilibrium is *fully active* if $q_k^*(\omega) > 0$ for all k and ω. In what follows, we restrict attention to fully active equilibria. This is an equilibrium-selection assumption and is not without loss of generality. However, the results in this appendix extend to equilibria with inactive platform–type pairs, with the usual complementary-slackness modifications. Since these corner cases introduce no new economic force, we suppress them here.

In a fully active equilibrium, every platform must deliver the same value to type-ω consumers. Hence, for every k and ω,

$$B_k^*(\omega) = M^*(\omega) = F_\omega^{-1}\left(\frac{Q^*(\omega)}{\bar{q}(\omega)}\right) \triangleq r^*(\omega).$$

Thus, $p_k^*(\omega) + U_k(\omega, x_k^*) = r^*(\omega)$ for all $k \in \mathcal{K}$ and $\omega \in \Omega$.

Efficiency. Fix an allocation (q, x) and let $Q(\omega) \triangleq \sum_{k \in \mathcal{K}} q_k(\omega)$. Given (q, x), the welfare of the platforms and the consumers is

$$W(q, x) = \sum_{k \in \mathcal{K}} \sum_{\omega, a_k} \left[v_k(a_k, \omega) + u_k(a_k, \omega)\right] x_k(a_k|\omega) q_k(\omega) + R(Q),$$

where $R(Q)$ is defined as in the main text.

Definition F.2. *An allocation (q, x) is efficient if it solves*

$$\begin{aligned} \max_{q_k \in \mathbb{R}_+^\Omega, x} \quad & W(q, x) \\ \text{s.t.} \quad & 0 \leq Q(\omega) \leq \bar{q}(\omega) \text{ for all } \omega \text{ and } x_k \in \mathcal{X}_k(q_k) \text{ for all } k. \end{aligned}$$

Note that this efficiency benchmark differs from the one in Section 3.1 because we evaluate efficiency only in terms of the welfare generated for platforms and consumers, thereby excluding merchants' payoffs. The reason is simple. In this more general specification of the model, platforms need not have full bargaining power vis-à-vis merchants and, therefore, need not fully internalize their profits. This may generate a downstream inefficiency that is unrelated to the data market and instead reflects a standard monopoly distortion. We therefore adopt a more restrictive efficiency benchmark that abstracts from these downstream distortions, allowing us to isolate the data-market distortions that are the focus of this paper.

As in the main text, we let the reduced welfare function be $\widehat{W}(q) \triangleq \max_{x_k \in \mathcal{X}_k(q_k)_{k\in\mathcal{K}}} W(q,x)$. Local efficiency is defined analogously as in the main text.

The Data Externality and Main Results. As in the main text, a platform database q_k is regular if $\mathcal{X}_k(q_k)$ is a singleton and V_k is differentiable at q_k. A database vector q is regular if q_k is regular for every k. Fix a fully active equilibrium and suppose that q^* is regular. For each k, there is a locally unique differentiable selection $q_k \mapsto x_{k,q_k} \in \mathcal{X}_k(q_k)$, with $x_{k,q_k^*} = x_k^*$. Along this selection, welfare can be written as

$$W(q,x_q) = \sum_{k\in\mathcal{K}} V_k(q_k) + \sum_{k\in\mathcal{K}} \sum_{\omega} q_k(\omega) U_k(\omega, x_{k,q_k}) + R(Q).$$

Differentiating with respect to $q_k(\omega)$ at q^* gives

$$\frac{\partial W(q,x_q)}{\partial q_k(\omega)}\Big|_{q=q^*} = \frac{\partial V_k(q_k^*)}{\partial q_k(\omega)} + U_k(\omega, x_k^*) - r^*(\omega) + \varepsilon_k^*(\omega),$$

where

$$\varepsilon_k^*(\omega) \triangleq \sum_{\omega', a_k} q_k^*(\omega') u_k(a_k, \omega') \frac{\partial x_{k,q_k}(a_k|\omega')}{\partial q_k(\omega)}\Big|_{q_k=q_k^*}.$$

This term, which is the analogue of Equation (4) in the main text, captures the effect of assigning a marginal ω-record to platform k on the consumers already participating in platform k's pool. It generates two possible distortions. If the record would otherwise not have been sold, the relevant externality is $\varepsilon_k^*(\omega)$. If instead the record would otherwise have been sold to platform k', the relevant externality is $\varepsilon_k^*(\omega) - \varepsilon_{k'}^*(\omega)$. The first is a participation externality; the second is an assignment externality. Thus, the multi-platform economy introduces additional margins of inefficiency, but their source remains the same data externality identified in the main text.

Proposition F.1 (Analogue of Proposition 1)**.** *Fix a fully active equilibrium $(p^*, q^*, x^*, \alpha^*)$ such that q^* is regular. The equilibrium is locally efficient if and only if $\varepsilon_k^*(\omega) = 0$ for all k and ω. If, in addition, $\widehat{W}$ is concave, the equilibrium is efficient.*

Proof. Since the equilibrium is fully active, $q_k^*(\omega) > 0$ for all k and ω. Since each F_ω has full support and all payoffs are finite, $0 < q_k^*(\omega) \le Q^*(\omega) < \bar{q}(\omega)$ for all k and ω. Hence every direction $d = (d_k)_{k\in\mathcal{K}} \in \mathbb{R}^{K|\Omega|}$ is locally feasible. By regularity, each platform-optimal joint allocation is locally affine on its linear-programming cell. Therefore, in a neighborhood of q^*, the reduced welfare function is the sum of an affine function of q and the term $R(Q)$, where

Q is linear in q. Since R is concave, $\widehat{W}$ is locally concave around q^*. Differentiating $\widehat{W}$ along the unique optimal selections x_{k,q_k} gives $D\widehat{W}(q^*)[d] = \sum_{k,\omega} d_k(\omega)\varepsilon_k^*(\omega)$. Hence q^* is a local maximizer of $\widehat{W}$ if and only if $D\widehat{W}(q^*)[d] \leq 0$ for every locally feasible direction d. Since $-d$ is also locally feasible, this holds if and only if all coefficients vanish, that is, if and only if $\varepsilon_k^*(\omega) = 0$ for all k and ω. If, in addition, $\widehat{W}$ is concave, the first-order condition is sufficient for global maximization. □

Proposition F.2 (Analogue of Proposition 2)**.** *Generically, every fully active equilibrium in which some platform uses a nondeterministic recommendation mechanism is inefficient.*[19]

Proof. Fix a fully active equilibrium $\left((p_k^*, q_k^*, x_k^*)_{k\in\mathcal{K}}, \alpha^*\right)$. Suppose that, for platform k, action profile a_k is recommended with positive probability. Let $q_k^{a_k}(\omega) \triangleq x_k^*(a_k|\omega)q_k^*(\omega)$. For small t of either sign, perturb only platform k's database by setting $q_{k,t} = q_k^* + tq_k^{a_k}$, and $q_{\ell,t} = q_\ell^*$ for $\ell \neq k$. By the same segment-scaling argument used in the proof of Proposition 2, applied to platform k's public-recommendation linear program, there exists $x_{k,t} \in \mathcal{X}_k(q_{k,t})$ with $x_{k,0} = x_k^*$; all other platforms' mechanisms are unchanged. The welfare derivative along this path is

$$\frac{d}{dt}W(q_t, x_t)\Big|_{t=0} = \sum_{\omega} q_k^{a_k}(\omega)\left[u_k(a_k, \omega) - U_k(\omega, x_k^*)\right].$$

Indeed, the derivative of platform k's continuation value is $p_k^* \cdot q_k^{a_k}$, the derivative of consumer surplus is $\sum_\omega q_k^{a_k}(\omega)u_k(a_k, \omega)$, and the derivative of the outside-option term is $-\sum_\omega q_k^{a_k}(\omega)r^*(\omega)$. Since the equilibrium is fully active, $r^*(\omega) = p_k^*(\omega) + U_k(\omega, x_k^*)$ for every ω, yielding the displayed formula. Local efficiency requires this derivative to vanish for every used public segment.

Suppose now that x_k^* is nondeterministic. Then there exist a_k and ω_0 such that $0 < x_k^*(a_k|\omega_0) < 1$. The displayed derivative is a linear expression in the consumer-payoff primitives u_k. Its coefficient on $u_k(a_k, \omega_0)$ is $x_k^*(a_k|\omega_0)q_k^*(\omega_0)\left[1 - x_k^*(a_k|\omega_0)\right]$, which is strictly positive. Hence local efficiency imposes a nontrivial linear restriction on the consumer-payoff primitives. The finite-stratification and Fubini argument used in the proof of Proposition 2 applies platform by platform. Therefore, generically, a fully active equilibrium with a nondeterministic platform mechanism cannot be locally efficient, and hence cannot be efficient. □

[19] The genericity concept is the same as that discussed for Proposition 2.

Proposition F.3 (Analogue of Proposition 3). *Fix a fully active equilibrium* $(p^*, q^*, x^*, \alpha^*)$ *such that* q^* *is regular. If every* x_k^* *is deterministic, then the equilibrium is locally efficient. If, in addition,* $\widehat{W}$ *is concave, the equilibrium is efficient.*

Proof. Since q^* is regular, each platform has a differentiable local optimal selection $q_k \mapsto x_{k,q_k} \in \mathcal{X}_k(q_k)$, with $x_{k,q_k^*} = x_k^*$. Fix k, a_k, and ω. The map $q_k \mapsto x_{k,q_k}(a_k|\omega)$ is differentiable at the interior point q_k^* and takes values in $[0, 1]$. Since x_k^* is deterministic, this coordinate is either 0 or 1 at q_k^*. If its derivative in some direction were nonzero, then moving a small amount in one of the two directions would push the coordinate outside $[0, 1]$. Hence $Dx_{k,q_k}(a_k|\omega)\big|_{q_k=q_k^*} = 0$ for all k, a_k, ω. By the definition of $\varepsilon_k^*(\omega)$, this implies $\varepsilon_k^*(\omega) = 0$ for all k and ω. Proposition F.1 then gives local efficiency. If $\widehat{W}$ is concave, local efficiency implies efficiency. □

Relationship to the Baseline Model. The model in the main text is a special case of the more general formulation in this appendix. In the baseline model, the platform can extract the merchants' entire continuation surplus through access fees. Hence the platform's effective payoff is its direct payoff v plus the merchants' payoff π. Under that specialization, Definition F.2 coincides with social welfare. Thus, the full-extraction assumption in the main text is a convenient way to shut down downstream contracting distortions and isolate the data externality. It is not essential for the externality identified in our main results.